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EvoAI enables extreme compression and reconstruction of the protein sequence space

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Designing proteins with improved functions requires a deep understanding of how sequence and function are related, a vast space that is hard to explore. The ability to efficiently compress this space by identifying functionally important features is extremely valuable. Here we establish a method called EvoScan to comprehensively segment and scan the high-fitness sequence space to obtain anchor points that capture its essential features, especially in high dimensions. Our approach is compatible with any biomolecular function that can be coupled to a transcriptional output. We then develop deep learning and large language models to accurately reconstruct the space from these anchors, allowing computational prediction of novel, highly fit sequences without prior homology-derived or structural information. We apply this hybrid experimental–computational method, which we call EvoAI, to a repressor protein and find that only 82 anchors are sufficient to compress the high-fitness sequence space with a compression ratio of 10^{48}. The extreme compressibility of the space informs both applied biomolecular design and understanding of natural evolution.

Protein engineering and design can create proteins with optimized functions for various applications in biotechnology, medicine and synthetic biology[1–3]. The fundamental challenge of protein engineering is to understand and manipulate the protein fitness landscape, which is a high-dimensional and complex space that contains a vast number of possible sequences and functions[4,5]. Although there have been considerable attempts over the past several decades to search this space for high-fitness sequences, we have only scratched the surface of understanding the rules and features of the space[6–9].

Experimental methods using directed evolution techniques, such as deep mutational scanning (DMS)[10,11], site-saturated mutagenesis[12,13] and random library construction[14,15], can provide valuable information, but they are laborious and time-consuming to scale up and typically must trade off accuracy and precision with sequence space coverage. These experimental methods are also usually restricted to low-dimensional mutations that do not take into account the natural selection pressure that shapes the protein fitness landscape in high-dimensional space. Advanced directed evolution tools that

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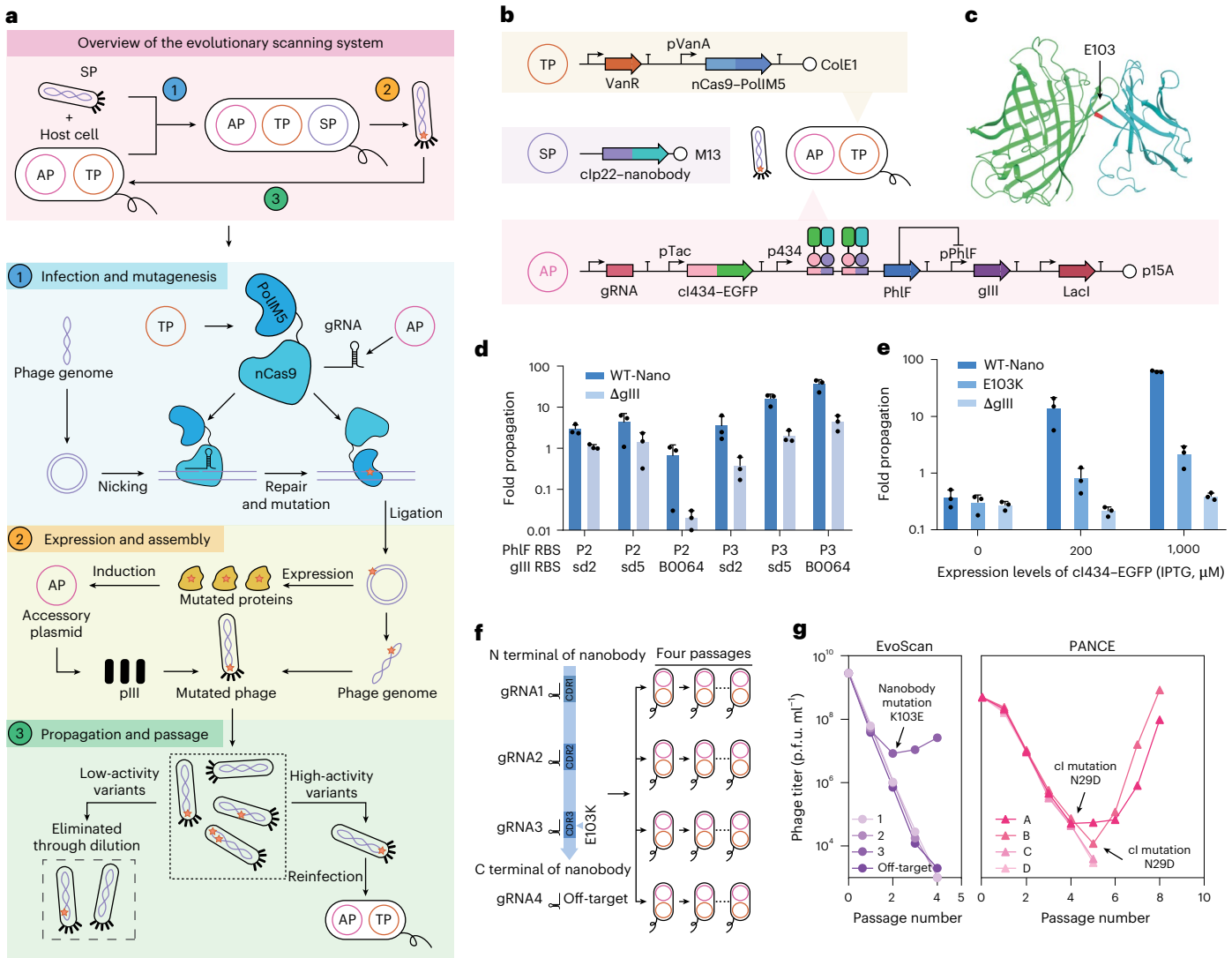


Fig. 1 | EvoScan scheme, development and validation on a protein–protein interaction evolution. a, Overview of the evolution scanning system. **b**, The testing scheme for the EvoScan system on EGFP–nanobody interaction. **c**, The predicted interaction between EGFP and nanobody. The structures of EGFP and nanobody were predicted by Alphafold2, and the interaction was modeled by ZDOCK (Methods). The position of the E103 site is labeled in red. **d**, Propagation assays of combinatorial AP designs. Several RBSs were tested, including P2 and P3 for PhlF and sd2, sd5 and B0064 for gIII. **e**, Propagation assays of WT-Nano phage, the E103K mutant phage and empty phage (ΔgIII) under different concentrations of IPTG (0, 200 μM and 1 mM) to control cI434–EGFP expression levels. EGFP, enhanced green fluorescence protein; WT-Nano, wild-type EGFP nanobody. **f**, EvoScan of the EGFP nanobody. Three different gRNAs targeted different CDR regions of the E103K nanobody. An off-target gRNA with no target sequence in the nanobody was used as a control group. CDR, complementary determining region. **g**, Phage propagation and mutations of EvoScan and PANCE for EGFP nanobody. Initial titers of phage E103K were 3×10^9 p.f.u. ml^{-1} for EvoScan and 5×10^8 p.f.u. ml^{-1} for PANCE. The dilution factor was 100 for each passage. Sequences of single guide RNA (sgRNA) and genetic parts are provided in Supplementary Tables 5 and 6. The data are mean ± standard deviation of three experiments, except for phage titers.

support the necessary scale, such as phage-assisted continuous evolution (PACE)[16,17] or OrthoRep[18], provide information primarily about trajectories that lead to high-fitness variants, which is insufficient to model the fitness landscape in its entirety. Computational methods, such as structure or sequence-based modeling of the protein fitness landscape[19–25], can evaluate larger sequence spaces but are limited by the availability and quality of training data, especially for proteins with few homologs or no structure information. These computational methods also typically do not account for other biological factors that affect the protein function, such as in vivo interactions or post-translational modifications.

An ideal approach to understanding and navigating this space for design and engineering purposes would use comprehensive high-throughput experimental data to inform efficient computational models. It was shown that high-throughput short sequencing data from directed evolution experiments can enable machine learning methods to reconstruct the full-length genotype and identify high-fitness variants[26]. Furthermore, it has been demonstrated that deep learning models for protein design can benefit from even a limited number of functionally characterized variants[27]. A recent work demonstrated that the protein fitness landscape is rugged with many local peaks but still easily navigable[28]. We view these functional variants or local peaks as key 'anchor' points that capture the features of high-fitness genotype space. We hypothesize that the design space for high-fitness genotypes can be effectively compressed by identifying a sufficient number of these anchor points to capture all the essential features, which can then instruct deep learning models to reconstruct and explore the whole space. However, no existing method can generate these anchors in a rapid and comprehensive way, especially for anchors from the high-dimensional space. Such a method would need

to capture functional information about variants evenly distributed across protein sequence space in a very high-throughput manner.

Here, we present EvoAI, an approach to empirically interrogate, then model, compress and reconstruct, the sequence space. Our approach combines high-throughput experimental evolution and computational methods to capture and learn from the essential features of the space. We first developed an evolutionary scanning method that adapts phage-assisted noncontinuous evolution (PANCE)[17] by incorporating a segmented mutagenesis system based on EvolvR[29]. Compared with traditional methods, this method enabled rapid and thorough evolutionary scanning from low to high dimensions and captured valuable fitness anchors. We then developed a deep learning and large language model to reconstruct the sequence space from these anchors and design new proteins, all ten of which showed greatly improved (up to 11 times higher) activity than wild type (WT). For a repressor protein, we demonstrated that this vast design space can be extremely compressed by a factor of 10^{48} to 82 points.

Results

The evolutionary scanning method

The M13 bacteriophage has a single-stranded DNA genome, but it generates a double-stranded form after infecting the host cell[30] (Fig. 1a). We reasoned that this should allow the EvolvR mutagenesis system that used CRISPR-guided DNA polymerase to introduce mutations into the M13 phage genome for selection and evolution[29]. Here, the expression of the nCas9–PolI complex was controlled by the vanillic acid-induced VanR–pVanA expression system that has a large induction fold change and low background expression and is suitable for expressing large and highly toxic proteins[31,32]. This system was incorporated into a medium-copy ColE1 plasmid to create the targeted mutagenesis plasmid (TP), replacing the mutagenesis plasmid (MP) used in traditional PACE. The evolution target was inserted into the M13 genome in place of gIII (the major coat protein of M13) to generate the selection phage (SP). The accessory plasmid (AP) expresses guide RNAs (gRNAs) that target different regions of the gene of interest for mutagenesis. The AP also contains gIII under the control of a genetic circuit that links the function of the gene of interest to the expression of gIII. This allows the selection of phages with improved and high-fitness protein function during phage propagation, while phages with nonfunctional genes are eliminated after dilution (Fig. 1a). We named this system EvoScan (evolutionary scanning).

EvoScan can explore specific regions of the fitness landscape to generate valuable anchors. These anchors are obtained by using different gRNAs to divide the target gene into defined segments, thus reducing the dimensionality of the fitness space. Moreover, the combination of different gRNAs through serial propagation on host cells bearing different APs enables the scanning and identification of anchors in higher dimensions, which can capture more details of the protein sequence space. To investigate and scan the protein sequence space, we validated and used this system to study three proteins with diverse functions: an enhanced green fluorescent protein (EGFP)-specific nanobody for protein–protein interaction; severe acute respiratory syndrome coronavirus 2 main protease (SARS-CoV-2 M^{pro}) and its inhibitors for protein–ligand interaction; and AmeR and its DNA operator for protein–nucleic acid interaction.

Validation of EvoScan and rapid identification of anchors in nanobody

To validate EvoScan and apply this system to proteins involved in protein–protein interaction, we chose antigen–antibody interaction, in this case, EGFP and its cognate nanobody[33]. We first established a reverse two-hybrid system that coupled the nanobody–EGFP interaction to the expression of gIII. We fused EGFP to the cI434 repressor, and its nanobody to cIp22, which can interact with cI434 but not with itself[34]. The gene encoding nanobody–cIp22 was inserted on phage to replace gIII. The gene encoding EGFP–cI434 was integrated on the AP and transformed into *Escherichia coli*. After phage infection, interaction between EGFP and nanobody will enable the interaction between cI434 and cIp22 to form a tetramer complex and inhibit the p434 promoter (Fig. 1b,c). In the AP, a transcriptional repressor PhlF was placed downstream of the p434 promoter, and gIII was placed under the control of the pPhlF promoter, such that interaction between EGFP and the nanobody will eventually induce the expression of gIII and allow phage propagation (Fig. 1b). We tested several combinations of ribosome binding sites (RBSs) and chose P3 RBS for PhlF and B0064 for gIII (Fig. 1d). This circuit propagated phages carrying EGFP nanobody while limiting the propagation of empty phages.

To test whether EvoScan could quickly identify a fitness-increasing protein variant anchor site, we artificially disrupted the interaction between EGFP and nanobody by introducing the E103K mutation in the complementary determining region 3 (CDR3) of the nanobody, which is essential for binding to its target (Fig. 1c,e). We designed four different gRNAs targeting different segments of the nanobody gene, with gRNA3 designed to target the segment containing the E103K mutation site of the nanobody (Fig. 1f). After two passages in EvoScan, we observed that only the group with gRNA3 targeting the E103K segment showed increased phage titer, while the other three groups all decreased (Fig. 1g). Sequencing results of the phage supernatant confirmed that, in the gRNA3 group, the E103K mutation had reverted back to glutamate. This validated that EvoScan can successfully and efficiently identify anchors that play important roles in protein function.

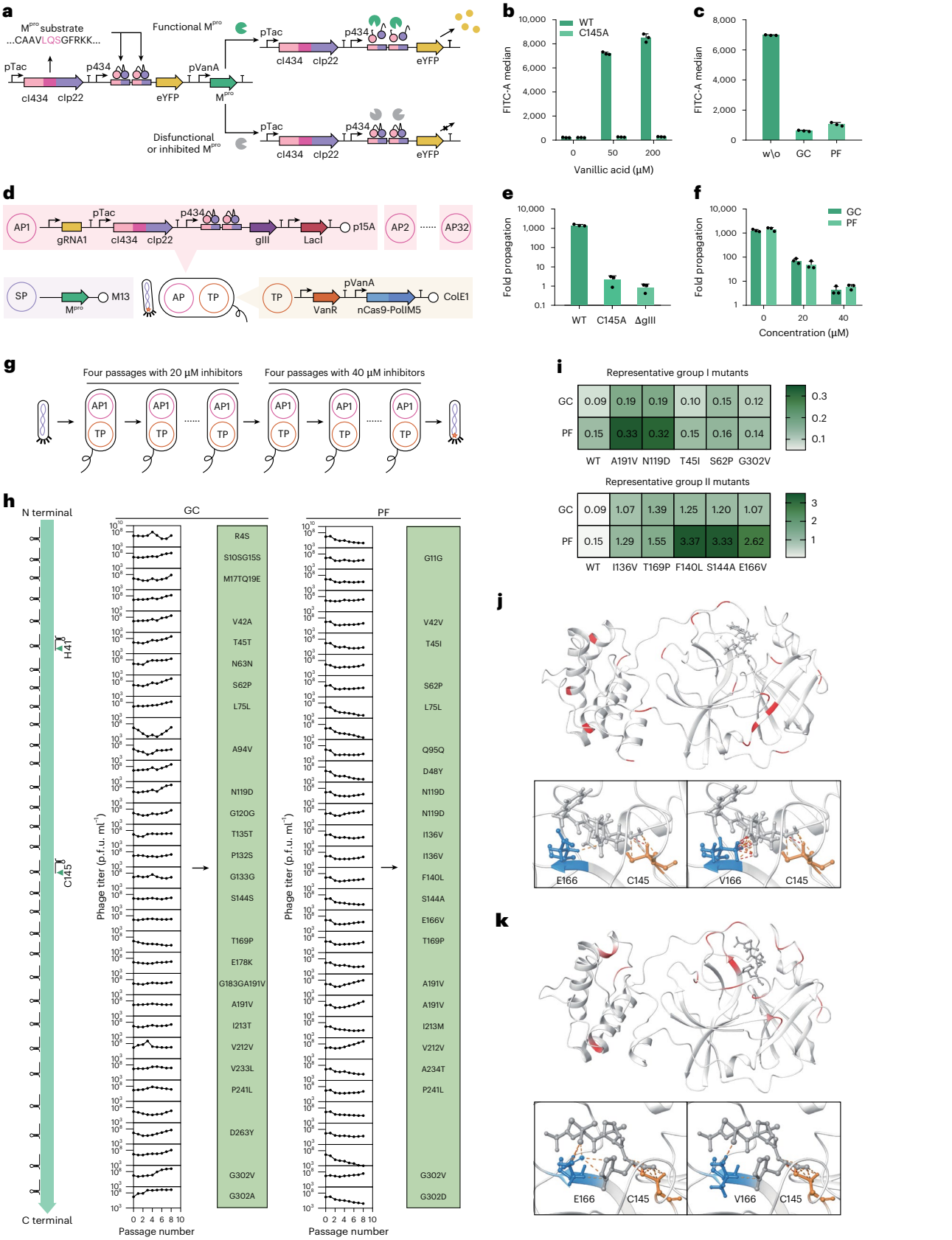
For comparison, we also implemented a traditional PANCE using the same E103K phage (Extended Data Fig. 1a). The two systems differed only in the use of targeted (EvoScan, TP) or nontargeted (PANCE, MP6) mutagenesis. After eight passages in PANCE, no consensus mutations were found in the nanobody gene. Interestingly, a N29D single mutation appeared on cIp22 (Fig. 1g and Extended Data Fig. 1b), which disrupted the selection pressure on nanobody function owing to the strong self-interaction between the two cI repressors (Extended Data Fig. 1c,d). These results further demonstrated that EvoScan can rapidly guide the evolution for precise searching of target proteins, even in the context of a more likely background mutation that could interfere with the desired evolution process.

Fig. 2 | Thorough segment scanning for protein–ligand interaction evolution. a, A schematic of the M^{pro} activity fluorescence reporter system. The M^{pro} substrate peptide CAAVLQSGFRKK was cloned into the linker between cI434 and cIp22 so that protease activity controlled the output (eYFP) from the p434 promoter. **b**, Flow cytometry assays for M^{pro} and the C145A mutant under different inducer concentrations (0, 50 and 200 μM vanillic acid and 200 μM IPTG). **c**, Flow cytometry assays for M^{pro} activity in the presence of 50 μM inhibitors (without inhibitor (w/o), GC376 (GC) and PF-07321332 (PF)). The concentration of IPTG was 200 μM, and the concentration of vanillic acid was 100 μM. **d**, The genetic circuit design for M^{pro} evolution. Thirty-two different APs carrying 32 different gRNAs tiling the M^{pro} gene sequence were designed. **e**, Phage propagation assays for M^{pro}, the C145A mutant and the empty phage at 1 mM IPTG. **f**, Phage propagation assays in the presence of 0, 20 or 40 μM inhibitors. **g**, A schematic diagram of the M^{pro} EvoScan process. **h**, EvoScan of M^{pro} using two different inhibitors and 32 different gRNAs. The initial titer of M^{pro} phage was 5×10^6 p.f.u. ml^{-1}. **i**, The RI against GC376 (50 μM) and PF-07321332 (50 μM) of different variants. **j**,**k**, The crystal structure of WT M^{pro} interacting with GC376 (**j**, Protein Data Bank (PDB) ID: 7CB7) and PF-07321332 (**k**, PDB ID: 7VLO). Mutated residue sites obtained in EvoScan are highlighted in red. Interactions between key residues and the ligand are indicated with dashed lines in the enlarged figures. Sequences of gRNAs and genetic parts are provided in Supplementary Tables 5 and 6. The data are mean ± standard deviation of three experiments, except for phage titers.

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Thorough identification of anchors reveals novel M^{pro} drug-resistant variants

We next applied EvoScan to investigate protein–ligand interaction. In this case, we chose M^{pro}, a crucial protease in the SARS-CoV-2 virus[35,36]. Several M^{pro} inhibitors have been developed and used to treat patients with coronavirus disease 2019, such as GC376[37] and PF-07321332[38], which is a key component of Paxlovid. However, the rapid mutation of SARS-CoV-2 may reduce or even eliminate the efficacy of these drugs.



Previous studies have identified mutational hotspots for drug resistance but have not comprehensively profiled the M^{pro} drug resistance fitness landscape[39,40]. It is important to thoroughly study possible escape mechanisms of M^{pro} in order to inform future drug development efforts. Here, we used EvoScan to systematically identify and extract key anchors from different regions of M^{pro} that affect its interaction with small molecule inhibitors.

To couple the protease activity of M^{pro} to the expression of downstream reporter genes, we fused the two cI repressors, cI434 and cIp22, with a linker that contains the specific sequence motif recognized by M^{pro}, such that only functional M^{pro} will cleave and deactivate the fused cI repressor (Fig. 2a). We also used a previously reported inactive M^{pro} mutant (C145A) to validate this system[35,36]. Our results demonstrated that this selection circuit can accurately and sensitively report on M^{pro} activity and the inhibition efficiency of small molecules (Fig. 2b,c). In addition, we found that this genetic circuit can be used for proteases from other viruses such as HCV (Extended Data Fig. 2a,b), demonstrating its robustness and broad applications. Compared with previously reported selections used for protease evolution in PACE[41–43], our circuit represents an alternative and improved strategy with better response properties (Extended Data Fig. 2c,d).

We then used our genetic circuit to couple the cleavage activity of M^{pro}, encoded on SP, to the expression of gIII, which was controlled by the p434 promoter (Fig. 2d). This circuit enables selection for M^{pro} variants that can escape inhibition by small molecules. Our results showed that WT M^{pro} supported robust phage propagation, while the C145A mutant behaved like empty phage (Fig. 2e and Extended Data Fig. 2e). We tested phage propagation at various concentrations of the inhibitors GC376 and PF-07321332, and selected 20 μM as the initial concentration for evolution (Fig. 2f and Extended Data Fig. 2f). We designed 32 different gRNAs to systematically cover the whole gene of M^{pro} and performed EvoScan with two inhibitors (Fig. 2g,h). Surprisingly, we found that escaping mutations can occur across the whole M^{pro} protein (Fig. 2h). Some of these mutations, such as F140L, E166V and S144A, have also been reported in previous studies on drug resistance against PF-07321332[39,44], proving the effectiveness and reliability of our system. Most other mutations were not observed in previous works, demonstrating that EvoScan can successfully identify novel key mutations. We also identified conserved mutation sites for both inhibitors, such as S62, L75, N119, S144, T169, A191, P241 and G302 (Fig. 2h). Interestingly, we observed that the phage propagation trajectories of the 32 segments targeted for mutagenesis varied during the evolution process, and more than 10 segments showed no overall enrichment during serial passaging, suggesting that mutations within each of these segments taken individually cannot enable drug resistance, which may serve as regions for future drug development studies (Fig. 2h).

We further verified the ability of these mutations to confer inhibition resistance (Fig. 2i and Extended Data Fig. 2g,h). Nearly all of these mutations showed increased resistance against inhibitors compared with WT M^{pro}. In group I mutations, we found that A191V had a strong resistance effect against both inhibitors, while N119D had a moderate resistance effect, and other mutations had relatively weak resistance effects on their specific inhibitors. Strikingly, we found a set of group II mutations (such as E166V), of which the enzyme activities were even improved by inhibitors (Fig. 2i). Similar to a previously reported mechanism where GC376 increased the catalytic activity of M^{pro} mutants[45], E166V has a different interaction with the inhibitors compared with WT, which may then improve the dimerization of M^{pro} and, thus, the enzyme activity (Fig. 2j,k). The same phenomenon was observed with other mutations such as I136V, T169P, F140L and S144A. However, how these mutation sites increase the enzyme activities when inhibitors were added is not clear, as they are located far from the active pocket.

As a comparison, we also evolved M^{pro} using the PANCE system (Extended Data Fig. 3a,b). With only the mutagenesis method changed, M^{pro} SP failed to accumulate any consensus mutations after 36 passages. After 96 passages, 4 dominant variants with escaping abilities emerged in the four groups in total (Extended Data Fig. 3c–f). All these variants have the N119D or A191V mutation, which appeared after only eight passages in EvoScan. These results further showed that EvoScan can effectively explore protein–ligand interaction and identify novel key anchor mutations related to small molecule interactions.

Systematic searching for anchors in high-dimensional space

Having demonstrated that EvoScan can rapidly and thoroughly explore the sequence space and generate more diverse functional variants than traditional methods, we next applied this approach to protein–nucleic acid interaction and systematically searched the space from low to high dimensions. We selected AmeR, a transcriptional regulator from the TetR family that plays important roles in many biological processes and synthetic biology[46,47]. AmeR has few known sequence homologs, making it challenging to use traditional methods to explore its sequence–function relationship, especially in high-dimensional space (Fig. 3a). We planned to first carry out a rapid scan of all gRNAs that cover the full sequence of AmeR, then select only those that generated enriched mutations for further use. Several different evolution routes could then be designed using the remaining APs. Serial passaging of phage across hosts containing different APs would identify anchors in high dimensions—that is, combining multiple mutations in different segments—that thoroughly and representatively sampled the AmeR sequence space (Fig. 3a).

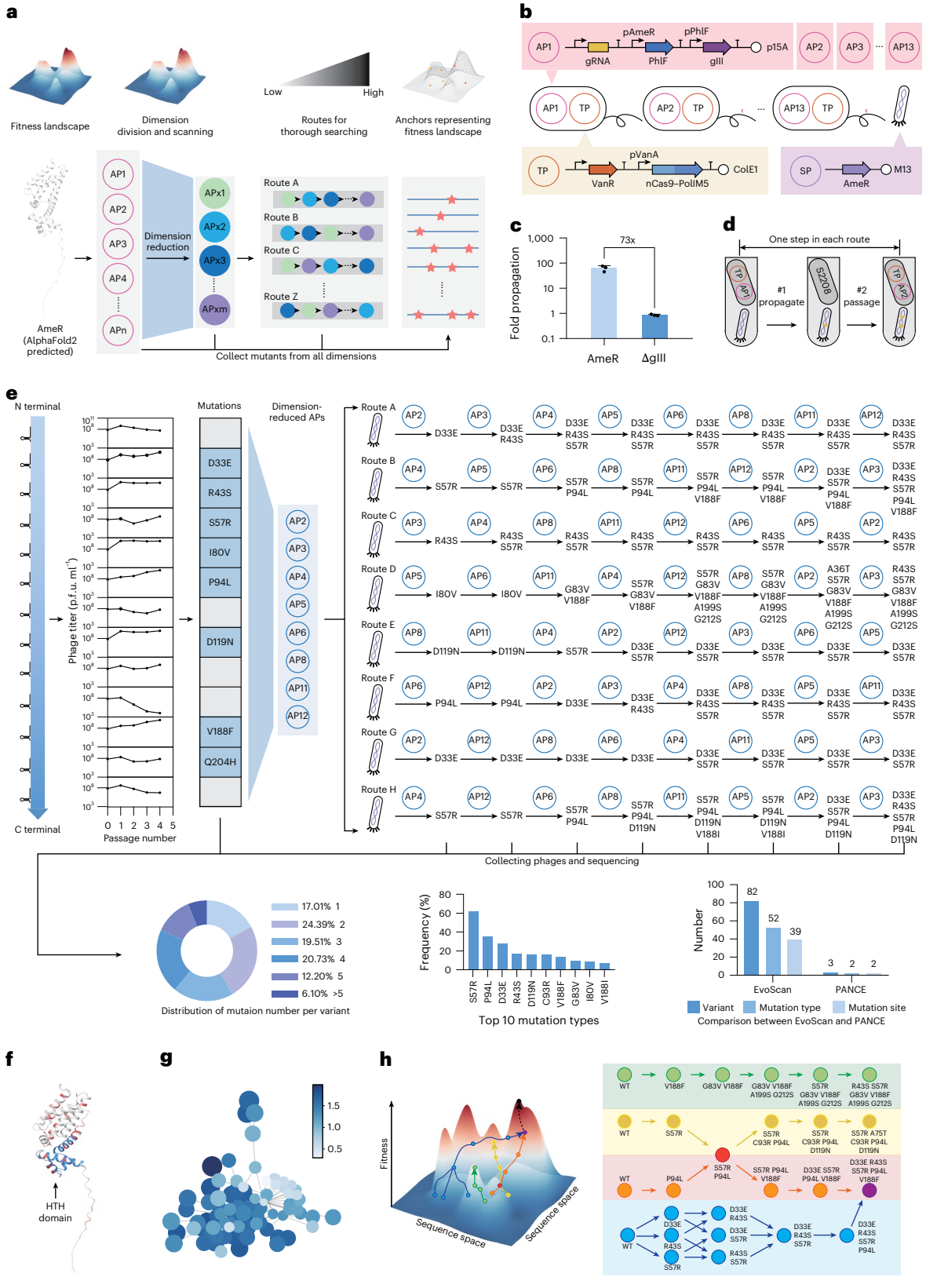
To link AmeR interaction with its operator to gIII expression, we inserted a PhlF repressor after the pAmeR promoter, such that the repression ability of AmeR is positively correlated with gIII expression (Fig. 3b). We tested several combinations of plasmid origins, RBSs and repressor types[48] to optimize the circuit. The optimal combination resulted in 73-fold propagation of SP carrying AmeR (Fig. 3c and Extended Data Fig. 4b).

To start the scanning process, we selected 13 gRNA sites that cover both the N-terminal and C-terminal domains of AmeR, which are involved in DNA binding and dimerization, respectively. We measured phage titers after each of the four passages and found that most groups enriched ≥50-fold. Of the 13 different groups, 8 generated dominant mutations in the phage supernatant and were selected for downstream scanning. The remaining groups generated either synonymous

Fig. 3 | Applying EvoScan on AmeR for protein–DNA interaction evolution. **a**, A schematic diagram of EvoScan of AmeR. A series of gRNAs was used to divide the AmeR gene into segments. An initial set of passages identified gRNAs that resulted in mutations, and several different evolution routes were designed with these APs. WT AmeR phages were passaged through these routes sequentially to scan and collect anchors. **b**, The genetic circuit design for AmeR evolution. Thirteen different APs carrying 13 different gRNAs were designed. **c**, Phage propagation assays of SP bearing the AmeR gene and the empty phage (ΔgIII). **d**, A schematic diagram of one step in each route during evolution. **e**, EvoScan of AmeR and properties of the collected variants. For each step in each route, the dominant mutations observed from supernatant are shown. The mutation number distribution and the top 10 mutation types of the 82 variants are shown. A comparison of the evolution results between EvoScan and PANCE is shown, including variant numbers, mutation diversities and mutated sites of AmeR. **f**, The distribution of mutations on the AlphaFold2-predicted structure of AmeR. The red regions are mutation sites, and the blue region is the typical helix–turn–helix (HTH) domain of TetR family proteins. **g**, The mutation relation map among the 82 variants and WT AmeR. Each circle is a variant, and its size is the mutation number. Variants are colored on the basis of their log(fold repression). Variants with less than three amino acid difference are linked together. **h**, Schematic and mutant information of four evolution paths from the mutation relation map. Sequences of gRNAs and genetic parts are provided in Supplementary Tables 5 and 6. The data are mean ± standard deviation of three experiments, except for phage titers.

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mutations or nondominant mutations when we sequenced single phage plaques and, thus, were not chosen for further exploration (Supplementary Table 1). Notably, all these mutations were observed within the targeting segment corresponding to each gRNA (Fig. 3e), further demonstrating the effectiveness of the target mutagenesis system. These results provide one-dimensional information about the protein sequence space. We next designed eight evolutionary routes to sample the high-dimensional space, in which SPs were passaged across all these eight APs in different orders (Fig. 3d,e). For each route, we sequenced the supernatant and two single plaques from each round (Fig. 3e). After the full evolutionary scanning process, we obtained 82 anchor variants encompassing 52 different mutations at



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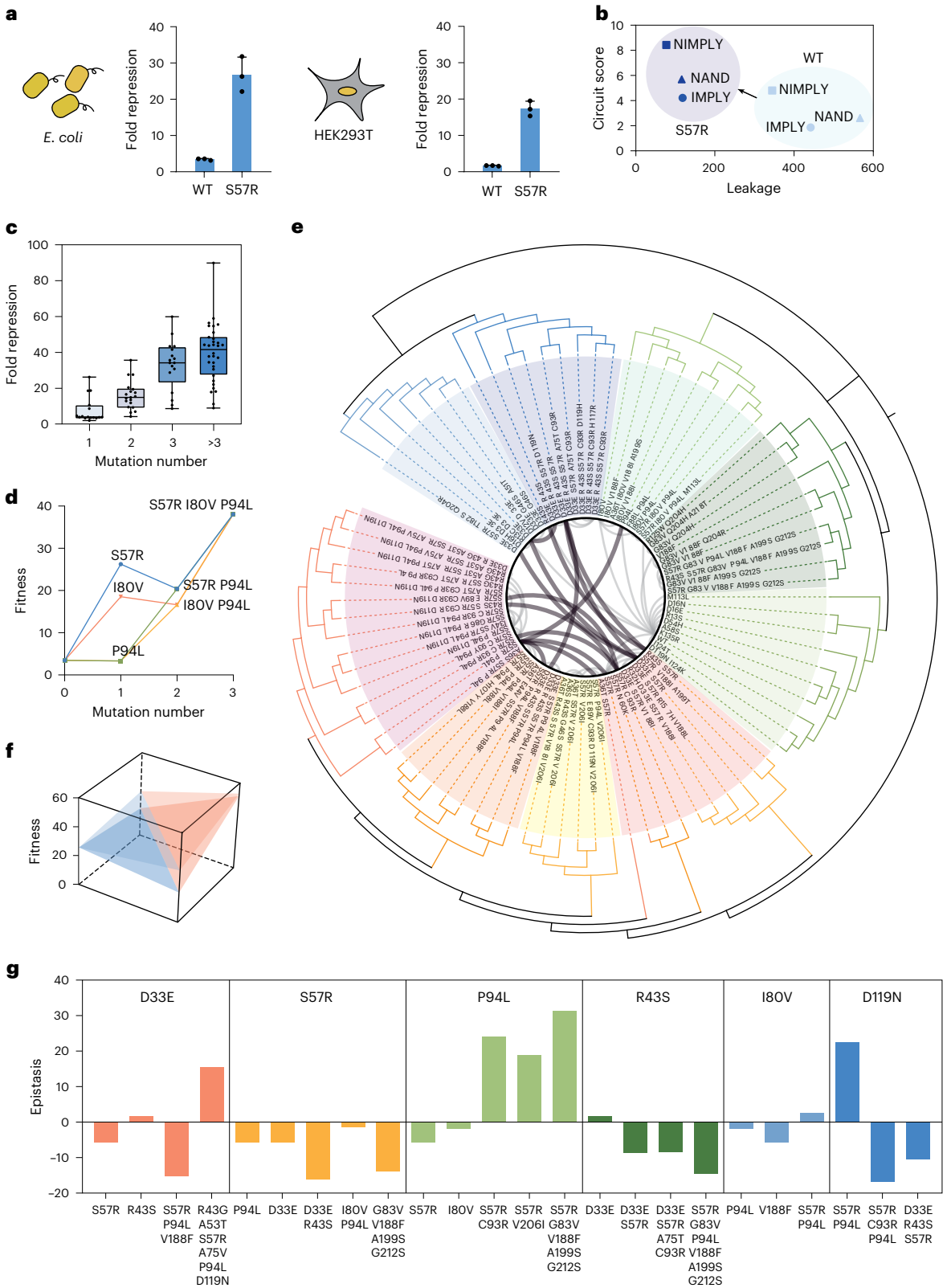


Fig. 4 | Genetic relationships and features of the 82 anchors generated by EvoScan. a, The fold repression of WT AmeR and the S57R variant in different systems, *E. coli* and HEK293T mammalian cells. **b**, Leakages and circuit scores of different genetic circuits (IMPLY, NIMPLY and NAND) using WT AmeR or the S57R variant. **c**, Fold repression of 82 AmeR variants with different mutation numbers. Each data point represents the average of three biological replicates. The center line represents the median value, while the box contains a quarter to three quarters of the dataset. The minima and maxima are also shown by the whiskers. **d**, Evolution paths from WT to the S57R I80V P94L variant. **e**, Phylogenetic tree of the 82 variants. The tree is divided into subregions by branch distances. Variants with less than two amino acid difference are linked by curves. Curves across subregions are in bold black. **f**, The magnitude epistasis for D33E and (P94L V188F) in the S57R genetic background (top plane) and reciprocal sign epistasis for P94L and (G83V V188F A199S G212S) in the S57R genetic background (bottom plane). **g**, Epistasis values of D33E, S57R, P94L, R43S, I80V and D119N with combinations of different mutations. The data are mean ± standard deviation of three experiments.

39 residue sites (Fig. 3e). Among all the variants, a large portion (~83%) of variants had more than two mutations, demonstrating the successful exploration and even sampling of the high-dimensional space. We measured the fold repression of the 82 variants, and nearly all of them showed improved function compared with WT, demonstrating again the effectiveness of EvoScan in searching for high-fitness sequences (Extended Data Fig. 5a and Supplementary Table 2).

For comparison, we also applied PANCE to AmeR evolution (Extended Data Fig. 4a–c). After 16 passages, only R43S and S57R single mutants and the R43S S57R double mutant appeared (Fig. 3e and Extended Data Fig. 4d,e), all of which appeared during EvoScan within eight passages. That only the variants from the low-dimensional space were observed in PANCE again illustrated how allowing competition between variants from all parts of sequence space can suppress and obscure many functionally informative mutation sites and high-fitness variants from the high-dimensional space, which were systematically captured by EvoScan.

Anchors capture key features of the design space

Alignment between mutations and predicted structure by AlphaFold2 suggested that these beneficial mutations accumulated not only on the helix–turn–helix domain near the N terminus that interacts with DNA, but also on regions related to dimerization of AmeR near the C terminus (Fig. 3f). To investigate the mutation relationship between variants, we drew a relation map linking variants that contained less than three different residues (Fig. 3g). The evolution paths leading to different mutants were connected with complexity, indicating the complex interactive nature of protein evolution in high-dimensional space.

We were able to identify four evolution paths from the complex map, leading to different mutants that shared the same intermediates or reached the same destination (Fig. 3h). This suggested the existence of shared local peaks in the landscape, consistent with a recent study demonstrating the simultaneous accessibility of multiple peaks during evolution[28]. These mutations usually contained one or more of D33E, R43S, S57R and P94L, and variants containing these mutations appeared to be fitter than WT AmeR (Fig. 3g and Extended Data Fig. 5a,b), indicating that these mutation sites provide important information about the sequence–function relationships for high-fitness genotypes.

The best-performing single mutant, S57R, outperformed the WT AmeR repression ability in both bacteria and mammalian cell systems (Fig. 4a and Extended Data Fig. 6a,b). Repressors with better properties are crucial for robust gates and genetic circuits construction in synthetic biology (for example, low leakage and high circuit score)[49]. We next incorporated it into several genetic circuit contexts such as IMPLY, NIMPLY and NAND[48] (Fig. 4b and Extended Data Fig. 6c–h). The S57R variant increased the circuit score of all these genetic circuits and reduced the circuit leakage at the same time (Fig. 4b). These results show that the identified mutations affected the protein–DNA interaction directly and captured essential features of the protein itself, rather than increasing fitness only in the context of our evolution selection.

Anchors capture complex epistasis interactions in the high-dimensional space

We also found that, in these anchors, mutational combinations had synergistically enhanced repression abilities in both *E. coli* and HEK293T, demonstrating that exploring the higher dimensions is vital for identifying proteins with improved functions (Fig. 4c and Extended Data Fig. 6a,b). However, we found that the order of introducing mutations affected the evolvability, even if the start point and end point were the same (Fig. 4d). For example, S57R P94L double mutants had lower fitness than S57R, which suggested that it was more difficult for natural evolution to reach the final genotype (I80V P94L S57R) if S57R was introduced first (Fig. 4d). We further built a phylogenetic tree to investigate the evolvability among these variants (Fig. 4e). The results revealed that, by designing different routes (Fig. 3e), EvoScan probably bypassed these evolvability limitations and achieved long genetic distance searching to obtain these anchors by 'jumping' between domains in different orders (APs) in the high-dimensional space. These results further highlighted the need for high-throughput targeting methods to effectively explore the sequence space.

These nonadditive interactions between two or more mutations are known as epistasis, which has profound impacts on the landscape in the high-dimensional space. We next systematically investigated the epistasis effect in these anchors and calculated the epistasis value (ε) using fold repression as the fitness value of different genotypes (Supplementary Table 3). We identified both negative (such as D33E and S57R, R43S and (D33E S57R A75T C93R)) and positive epistasis (such as (S57R P94L) and V188F, (P94L S57R) and (G83V V188F A199S G212S)) for different mutation combinations in both low dimensions and high dimensions (Supplementary Table 3). We also studied the magnitude and sign epistasis of different mutations (Fig. 4f), which can create rugged fitness landscapes. Interestingly, we identified reciprocal sign epistasis in the high-dimensional space, such as P94L and (G83V V188F A199S G212S) in the S57R genetic background (Fig. 4f). We also found that, even for the same mutation, such as D33E, P94L and D119N, ε can be either positive or negative when combined with different mutations, indicating the complex and idiosyncratic epistasis relationship between different mutations (Fig. 4g).

EvoAI enables sequence space reconstruction and prediction of new proteins

Given the complex interaction of mutations in the high-dimensional space, we next aimed to use deep learning to extract the latent features of these anchors obtained from EvoScan to represent and reconstruct the design space of AmeR for high-fitness genotypes with high accuracy, enabling the design of new proteins with multiple mutations not represented in the experimental outcomes. We name this hybrid experimental–computational method EvoAI.

We combined a pretrained GeoFitness model and the protein language model ESM-2, followed by a multilayer perceptron to enhance the accuracy of predicting protein mutation effects (Fig. 5a). The pretrained GeoFitness model was trained on a large dataset of ~300,000 protein fitness values from various experimental cases and indicators to enable prediction of protein fitness of single mutations (Extended Data Fig. 7). We used the 82 anchor points for both training and validation with a 10-fold cross-validation approach to obtain the final model (Extended Data Fig. 8). Spearman correlation coefficients were 0.91 and 0.84 for the training set and the test set, respectively, demonstrating a high level of consistency in training effectiveness (Fig. 5b). These results demonstrated that our deep learning model accurately predicted the multiinteraction of mutations and complex epistasis in higher-dimensional space.

We further validated the accuracy of the reconstructed space by designing, predicting and testing new variants different from the 82 anchors. To reduce the computational load, we chose 13 mutations from the top 11 mutation sites with high prediction certainties for novel protein design (Extended Data Fig. 9a,b). We then computationally traversed all possible combinations of 6 total mutations and calculated the predicted fold repression by our model (1,093 predictions in total). The ten top-scoring protein sequences were cloned and experimentally tested for their fold repression. All ten sequences showed improved activities compared with WT with 10- to 38-fold repression abilities (Supplementary Table 4). Furthermore, although we chose only the top predictions and all of them have very close prediction scores, these variants still showed a high Spearman correlation coefficient between prediction scores and experimental results (Extended Data Fig. 9c,d). For comparison, we tested the predicted sequence space without using these anchors information but only using low-dimensional DMS information, and also generated ten variants with six mutations each

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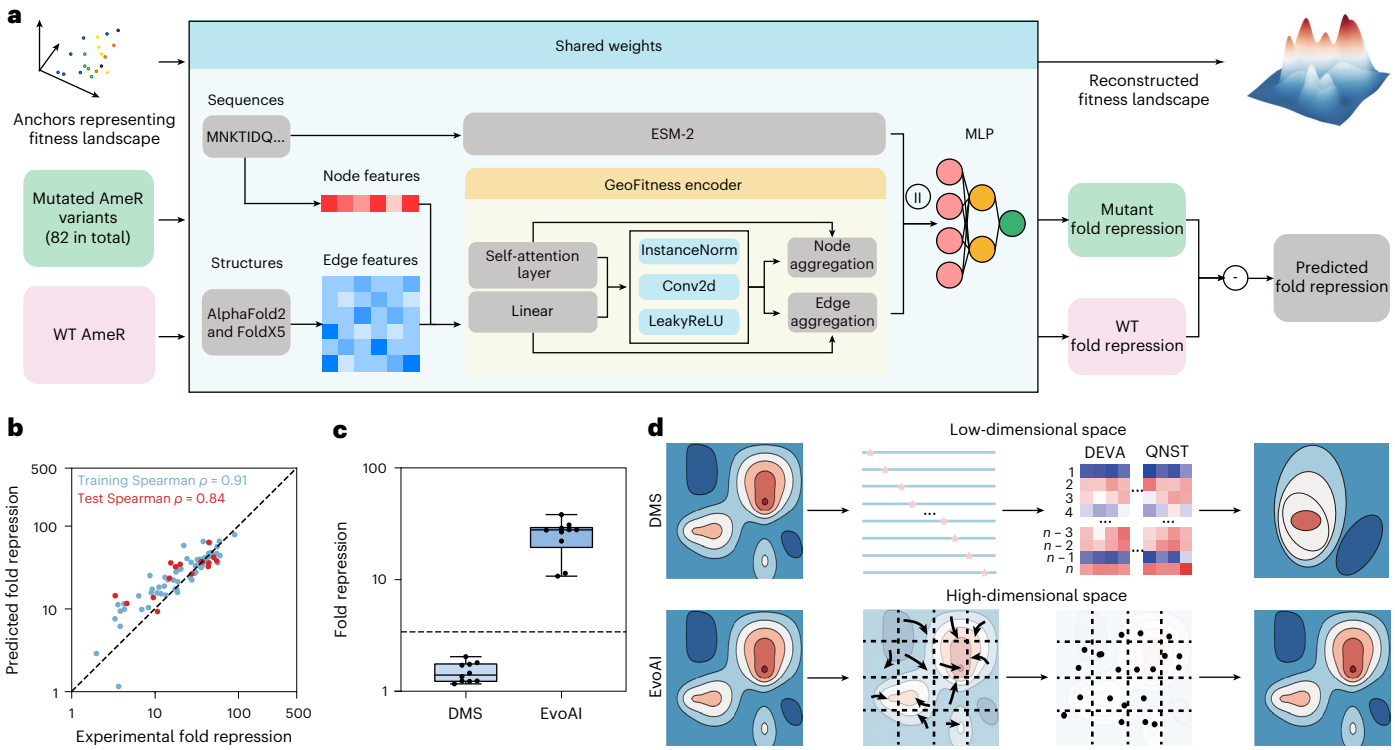


Fig. 5 | Anchors and deep learning reconstruct the design space for high-fitness genotypes. a, A schematic of the deep learning model. WT AmeR and 82 anchors from EvoScan were the dataset for model training. **b**, Training and test results of the 82 variants. Training data are in blue, and test data are in red. **c**, The experimental fold repression of the designed variants using model trained by EvoScan anchors (10, right) or DMS information (10, left). The dashed line is WT AmeR fold repression. The data points are the mean of three biological replicates. The center line represents the median value, while the box contains a quarter to three quarters of the dataset. The minima and maxima are also shown by the whiskers. **d**, A comparison between DMS and EvoAI. DMS can only search the low-dimensional space, while EvoScan comprehensively segments and scans the high-dimensional space to enable accurate design space reconstruction by deep learning.

(Extended Data Fig. 9e). In striking contrast to the high-performing EvoAI-predicted variants, all ten variants generated by DMS had worse activity relative to WT AmeR (Fig. 5c and Supplementary Table 4). We further constructed and measured ten intermediate and ten bottom variants predicted by EvoAI, and they showed fold repression consistent with the prediction (Extended Data Fig. 9f–h). Notably, all these variants outperformed the DMS group (Extended Data Fig. 9f), demonstrating the robust performance of our model for predicting high-fitness sequences.

These results validated that, with these compressed anchors, our deep learning model can accurately reconstruct the design space for high-fitness genotypes in high-dimensional space, and design new protein sequences with improved functions. We identified 39 mutation sites in AmeR (Fig. 3e) that could potentially generate high-fitness genotypes, with a theoretical design space of ~10^{50} (20^{39}). Our EvoAI approach therefore effectively demonstrated that this vast design space of AmeR for high-fitness genotypes can be compressed by ~10^{48} times to 82 anchor points.

Discussion

Navigating the complexity and scope of a protein fitness landscape is a long-standing challenge for protein design. We developed EvoScan, a system that combines EvolvR mutagenesis and phage selection to explore the protein sequence space in different dimensions. EvoScan can identify valuable anchors, which are variants with critical mutations that represent the sequence space. We showed that these anchor points can accurately reconstruct the space and design new proteins when coupled to deep learning methods (EvoAI), demonstrating the extreme compressibility of this space. Previous methods did not capture this insight probably because they only explored either the low-dimensional space by measuring single or double mutations, or a small region of the sequence space by saturating mutations. These methods thus might not capture the whole picture, especially the high-dimensional space (Fig. 5d).

Our approach has several important advantages over existing methods. First, it balances realistic fitness optimization and even sampling of sequence space, which can rapidly explore high dimensions, generate more diverse and functional variants, and provide richer information about sequence–function relationships. Second, by integrating empirical evolutionary scanning and deep learning models in EvoAI, we can leverage the strengths of both approaches. We could use the properties learned by deep learning to dynamically guide the scanning process. Future advances of explainable deep learning could uncover the underlying rules or patterns and provide insights into how proteins adapt and overcome evolutionary constraints or trade-offs. Third, our method can evolve and investigate proteins that lack structural information or that involve challenging interactions. We showed that EvoScan can capture anchors for proteins with diverse functions, such as protein–protein, protein–ligand and protein–nucleic acid interactions. Our approach should be compatible with any biomolecular function that can be coupled to a transcriptional output (for example, enzymes through small molecule sensors) and, thus, could be applied to study the sequence spaces of diverse biomolecules. However, it is worth noting that the chemical space, such as enzyme reactions, or the ligand space, such as protein interactions tested in our study, might differ and will require further exploration.

Our approach could be further improved in the future. We could use Cas9 variants with more protospacer adjacent motif options to

increase the gRNA tiling and mutation-targeted segment selection. We could also modify the editing system to introduce mutations at multiple sites at once, avoiding host switching and speeding up the exploration process. Furthermore, incorporation of the target mutagenesis approach of EvoScan into PACE could potentially lead to deeper sampling of sequence space segments. In addition, integration of EvoScan with genotype reconstruction methods, such as Evoracle, could enable more systematic and intelligent exploration of the sequence space[26]. Moreover, the modularity of our system makes it highly suitable for automation, such as with the recently reported PRANCE method[50], and could be scaled up to provide more comprehensive fitness landscape profiling data for different protein targets, illustrating whether the extreme compressibility of the design space for high-fitness genotypes is universal or unusual, or if the whole protein fitness landscape is compressible.

We also hope that our method will inspire new insights into the relationship between genotype and phenotype and the evolution of biological systems. The compressibility of the design space may suggest that nature somehow finds a way to search through the seemingly infinite space in the relatively short period of life time on earth by Darwinian evolution. Genetic recombination of chromosome regions in large sexual populations could possibly enable this dimension compression to boost evolution[51,52]. However, it is important to recognize that natural evolution is multifaceted, influenced by varying selection pressures, fluctuating environmental conditions, conditional neutrality and genomic contexts—factors not fully accounted for in our current approach. Our approach might enable the investigation of such path dependence of evolutionary outcomes of biological systems when combined with high-throughput experiments in the future[50,53] and provide valuable insights for evolution and protein design in biotechnology and biomedical applications.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41592-024-02504-2.

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Methods

General methods

The following working concentrations of antibiotics were used: carbenicillin (Solarbio, 50 μg ml^{-1}), kanamycin (Solarbio, 50 μg ml^{-1}), spectinomycin (Macklin, 50 μg ml^{-1}) and chloramphenicol (Macklin, 25 μg ml^{-1}). PHANTA 2× mix (Vazyme) was used for cloning PCR, and Flash 2× mix (Vazyme) was used for verification PCR and Sanger sequencing (Tsingke Bioscience). All cloning fragments were assembled by Golden Gate assembly (New England Biolabs) or ClonExpress assembly (Vazyme) methods. Plasmids were cloned in DH5α competent cells (HT Health). Synthetic genes were ordered from Tsingke Bioscience. Cloned plasmids were extracted by Tiangen DNA extraction kit. *E. coli* strain S2060[54] was used in all aspects of the EvoScan process, including system construction, evolution and plaque assays. The DH5α strain was used for flow cytometry experiments. Detailed information on the plasmids and SP used in this work is given in Supplementary Table 7. Statistical figures were generated by GraphPad Prism 8 or Python 3.7 scripts.

Phage propagation assay

Competent S2060 cells were transformed with corresponding AP in each experiment. Overnight cultures of single colonies inoculated in lysogeny broth (LB) medium with proper antibiotics were diluted 50 or 100 times and grown at 37 °C in 220 rpm shaker (ZQZY-B8, cultured in shake tubes, 5 ml system) or 1,000 rpm shaker (HUXI HW-400TG, cultured in 96-deep-well plate, 500 μl system) to log phase (OD_{600} ~0.4–0.6). These cells were then infected with SP at an initial titer of 5×10^6 plaque-forming units (p.f.u.) per milliliter. The mixture was further cultured overnight (16–20 h) at 37 °C in the shakers as described above and was centrifuged at 2,500*g* for 10 min. Phages in the supernatant was filtered by 0.22 μm bacterial filter and stored at 4 °C for further use.

Plaque assay

A single colony of chemically competent S2208 cells[54] (S2060 cells transformed with plasmid pJC175e) was cultured overnight in LB medium added with proper antibiotics. The saturated bacteria culture was diluted 50 or 100 times into LB medium with proper antibiotics and grown at 37 °C in 220 rpm shaker to log phase (optical density (OD_{600}) ~0.4–0.8) before use. Phages were serially diluted six to eight times with a dilution ratio of tenfold in each step in LB medium. Then, 10 μl of each phage dilution was mixed with 45 μl S2208 cells, and then 180 μl of liquid (50–65 °C) soft agar (LB medium and 0.5% agar) supplemented with 60 μg ml^{-1} Bluo-gal (Inalco S.p.A.) was added and mixed by pipetting. The whole mixture was immediately added onto 500 μl of bottom agar (LB medium and 1.5% agar) supplemented with 200 μg ml^{-1} Bluo-gal (Inalco S.p.A.) previously prepared in a 24-well plate. Then, the plates were incubated in 37 °C for overnight growth (14–18 h).

Calculation of fold propagation

For fold propagation measurement of the SP, initial phage titer and final phage titer were measured by plaque assays. We defined the ratio of final phage titer versus initial phage titer as the fold propagation of the phage.

Basic process of evolutionary scanning (EvoScan)

TP was first transformed to chemically competent S2060 cells, and then the prepared S2060-TP cells were used to prepare super chemically competent cell by Inoue method[55]. Chemically competent S2060-TP cells were transformed with corresponding APs. The resulting S2060-TP-AP bacteria were cultured overnight and diluted 50–100 times into 500 μl LB medium with antibiotics and inducers, and grown in 37 °C 1,000 rpm shaker to OD ~0.5. The phage titer for the first infection was around 5×10^6 to 5×10^8 p.f.u. ml^{-1}, and for the following passages the phages were subjected to a 1:50 or 1:100 dilution. Vanillic acid (Sigma-Aldrich, ethanol dissolved) at a final concentration of 50 μM was added to induce the expression of nCas9–PolIM5 complex. The mixture was then cultured in 37 °C 1,000 rpm shaker overnight. The next day, the mixture was centrifuged at 2,500*g* for 10 min and the phage content of the collected supernatant was verified by PCR (Flash 2× mix) and Sanger sequencing. The supernatant was then used for plaque assay as described above. Single plaques from plaque assay were picked and further verified by PCR (Flash 2× mix). The PCR product was sent for Sanger sequencing.

Searching steps in each route

For each step of EvoScan in a route, 10 μl supernatant with evolved phages was added into 1 ml log-phase S2208 bacteria culture (OD ~0.4–0.8) and propagated overnight in a 96-deep-well plate. The mixture was centrifuged at 2,500*g* for 10 min and filtered by 0.22 μm bacterial filter. The obtained phages were then diluted and infected another host cell containing a different AP with an infection titer of 5×10^6 p.f.u. ml^{-1}.

Basic process of PANCE

The AP with the designed genetic circuit and the mutagenesis plasmid MP6 were cotransformed into super chemically competent S2060 cells. The S2060-MP6-AP bacteria were cultured overnight and diluted 50–100 times into 500 μl LB medium with antibiotics and inducers in a 96-deep-well plate, and grown in 37 °C 1,000 rpm shaker to OD ~0.5. The initial phage titer was around 5×10^6 to 5×10^8 p.f.u. ml^{-1}, and the phages were subjected to a 1:10–1:100 dilution in the following passages in a 500 μl system. Then, 1% (m/v) arabinose dissolved in ddH_2O was added as the inducer of MP6, after which phages were collected to obtain mutations following the same procedures in EvoScan.

Induced expression assay

Single colonies of strains to be tested were cultured in LB medium overnight. Saturated bacterial culture was diluted 100 times in LB medium with proper antibiotics and inducers, and cultured in 37 °C 1,000 rpm shaker for 2 h (OD ~0.4). Then, LB with proper antibiotics and inducers was prepared and 2 μl log-phase bacteria culture was added together to a whole volume of 500 μl. The mixture was cultured for 5 h in the 96-deep-well plate.

Flow cytometry assay

Ten microliters of the culture was added into 190 μl phosphate-buffered saline with 2 g l^{-1} kanamycin in a 96-well U-bottom plate to stop the cell growth. The plate was stored at 4 °C until use. A flow cytometer (Beckman Coulter Cytoflex S) was used to quantify the expression levels of fluorescent protein. The software FlowJo v10 was used to gate the events (at least 10,000 events) and calculate the median of each sample.

M^{pro} drug resistance index

In the reverse two-hybrid system protease activity assay, the fluorescence of the experimental group carrying enhanced yellow fluorescence protein (eYFP) was measured with or without addition of M^{pro} inhibitor GC376 or PF-07321332. The ratio of the median value of fluorescence through the channel fluorescein isothiocyanate A (FITC-A) with inhibitor versus fluorescence FITC-A median without inhibitor was defined as the resistance index (RI) to evaluate the drug resistance abilities of different M^{pro} variants.

Structure display and interaction prediction

Schrodinger 2017 was used for structural display. ZDOCK[56] was used for interaction structure prediction between EGFP and its nanobody. The interaction between M^{pro} and inhibitors within 3 Å is shown in Fig. 2j,k.

Fold repression calculation

The background fluorescence of cells, which is the median of the fluorescence of the bacteria carrying an empty plasmid with only the

backbone, was measured and subtracted from all the experimental groups. The subtracted fluorescence values of the uninduced group (no repressor expression) were divided by the induced group (repressor expression) to obtain the fold repression.

Relative expression level calculation

Using flow cytometry assay, we measured the FITC-A median of the strain carrying the empty plasmid and set this value as the background value. The FITC-A median of the strain carrying the standard plasmid expressing eYFP through the open reading frame J23101-B0064-YFP was measured the same way and set as the standard value. The FITC-A median of the strain containing a specific variant was measured the same way, and the relative expression level was defined as (variant value − background value)/standard value.

Circuit score calculation

The strain carrying the plasmid with a specific genetic circuit was prepared for flow cytometry assay. Isopropyl β-D-1-thiogalactopyranoside (IPTG, 1 mM) and vanillic acid (100 μM) were used as the input signals. YFP was used as the output reporter of the circuit, and the FITC-A median of each state was measured. The lowest ON signal (lowest FITC-A median in 'ON' states of the circuit) was divided by the highest OFF signal (highest FITC-A median in 'OFF' states of the circuit) to obtain the circuit score.

AmeR phylogenetic tree construction

Protein sequences of the 82 variants and the WT were collected as a fasta file, and the file was input into MEGA11 for multiple sequence alignment (MSA)[57]. After MSA and phylogenetic analysis, neighbor-joining tree was selected as the method of tree construction. The output tree was decorated by iTOL (v6.9.1)[58], and all the parameters were set using default values.

Epistasis calculation

Epistasis between two different mutations, A and B, could be calculated as $\varepsilon = f_{ab} + f_{AB} - f_{Ab} - f_{aB}$, where f is the fitness of WT, double-mutant and single-mutant genotypes, respectively. $\varepsilon > 0$ means positive epistasis, while $\varepsilon < 0$ means negative epistasis.

Mammalian cell culture and transfections

HEK293T cells (CRL-3216, ATCC) were cultured in Dulbecco's modified Eagle medium (DMEM, Gibco) supplemented with 10% (v/v) fetal bovine serum (Biological Industries) and 1% (v/v) penicillin–streptomycin solution (Beyotime) at 37 °C, 100% humidity and 5% CO_2. In transfection experiments, 60,000–80,000 HEK293T cells in 0.2 ml of DMEM complete medium were seeded into each well of 48-well plastic plates (NEST) and grown for ~24 h. M5 HiPer Lipo2000 Transfection Reagent (Lipo2000, Mei5bio) was used in all transfection experiments following the manufacturer's protocol. Briefly, a sample mixture was prepared by mixing 150 ng repressor plasmid or 150 ng control plasmid (repressor deficiency) with 150 ng reporter plasmid in 0.7 μl Lipo2000. The mixture was incubated at room temperature for 20 min before adding to cells. Transfections were supplemented with 0.2 ml DMEM complete medium 24 h post-transfection. Cells were cultured for 2 days post-transfection before flow cytometry analysis.

Mammalian cell flow cytometry assay

Cells were trypsinized 48 h after transfection and were then centrifuged at 2,500*g* for 10 min at room temperature. The supernatant was removed, and the cells were resuspended in 1× phosphate-buffered saline. Fluorescence values were measured with a Cytoflex flow cytometer (Beckman Coulter). PB450-A and ECD-A channels were chosen for blue fluorescence protein (BFP) and mCherry measurement, respectively. Data were processed using FlowJo (TreeStar), gated by the area of the forward scatter and the side scatter (FSC-A/SSC-A), and then cell populations were selected by gating out the background BFP signal of untransfected cells to obtain the median of fluorescence. The median of fluorescence was calculated for >20,000 transfected cells for each sample. To reduce expression noises between samples, the mCherry:BFP fluorescence ratio was used to report the repressor activity[59]. The mCherry:BFP fluorescence ratio was calculated by $(\text{mCherry} - \text{mCherry}_0)/(\text{BFP} - \text{BFP}_0)$, where mCherry_0 and BFP_0 were the fluorescence values from untransfected HEK293T cells. The fold repression was calculated by $(\text{mCherry:BFP})_{\text{unrepressed}}/(\text{mCherry:BFP})_{\text{repressed}}$, where $(\text{mCherry:BFP})_{\text{unrepressed}}$ and $(\text{mCherry: BFP})_{\text{repressed}}$ were the fluorescence values of the states cotransfected with control plasmid or repressor plasmid.

Feature generation

Our initial step entails querying the UniRef30_2021_03 and bfd MSA databases. Subsequently, we employ AlphaFold2 to construct the structural representation of the WT protein. For this endeavor, we deploy the GeoFitness-Seq variant of the pretraining model. In the case of mutated proteins, structural configurations are generated using FoldX 5. The sequence features are extracted from the large-scale protein language model ESM-2, for the purpose of capturing global context information. Consequently, each node in the Geometric Encoder is initialized by the embedding of the corresponding residue derived from the ESM-2. Unlike conventional methodologies that rely upon interresidue distances and contacts to establish edges, each edge in the Geometric Encoder is initialized by the relative geometric relationship between a pair of residues derived from the protein three-dimensional structure[60].

Cross-validation

We employed a 10-fold cross-validation approach to find the hyperparameters of the model. The dataset, comprising 82 mutational data points, was divided into three parts: a training set (59 samples), a validation set (7 samples) and a test set (16 samples). Model evaluation was performed using the Spearman correlation coefficient (ρ) as the primary assessment metric.

Model training details

The model employs the soft rank loss as its loss function, with a learning rate of 10^{-3}, Adam optimizer and a decay rate for the learning rate. The training spans across 50 epochs. Subsequently, the learning rate of the upstream GeoFitness model is set to 10^{-4}, while the learning rate of the downstream model is adjusted to 5×10^{-4} for further fine-tuning.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Source data are provided with this paper. Other data and materials used in this study are available from the corresponding authors by reasonable request. PDB files used in this study include PDB ID 7CB7 and PDB ID 7VLO. All the prediction features and results of mutants are available via Zenodo at https://doi.org/10.5281/zenodo.10686156 (ref. 61).

Code availability

The models of EvoAI are implemented in PyTorch v2.2.0. All codes are freely downloadable via Zenodo at https://doi.org/10.5281/zenodo.10686156 (ref. 61) or via GitHub at https://github.com/Gonglab-THU/EvoAI.

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Author contributions

S.Z. conceptualized and supervised the project. S.Z., Z.M. and W.L. designed the experiments. Z.M., W.L. and H.Q. performed the evolution experiments in EvoScan. Z.M., W.L., Y.S. and Z.L. performed the flow cytometry assays and phage propagation assays of obtained variants. G.L. conducted the mammalian cell experiments. H.G., B.T., Y.X. and J.C. designed and developed the deep learning models. Z.M. and Y.S. wrote the first draft. B.W.T., D.R.L., C.A.V. and S.Z. wrote the final manuscript. All authors contributed to the drafting and revision of the manuscript.

Competing interests

S.Z. and Z.M. have filed a patent application based on this work. The other authors declare no competing interests.

Additional information

Extended data is available for this paper at https://doi.org/10.1038/s41592-024-02504-2.

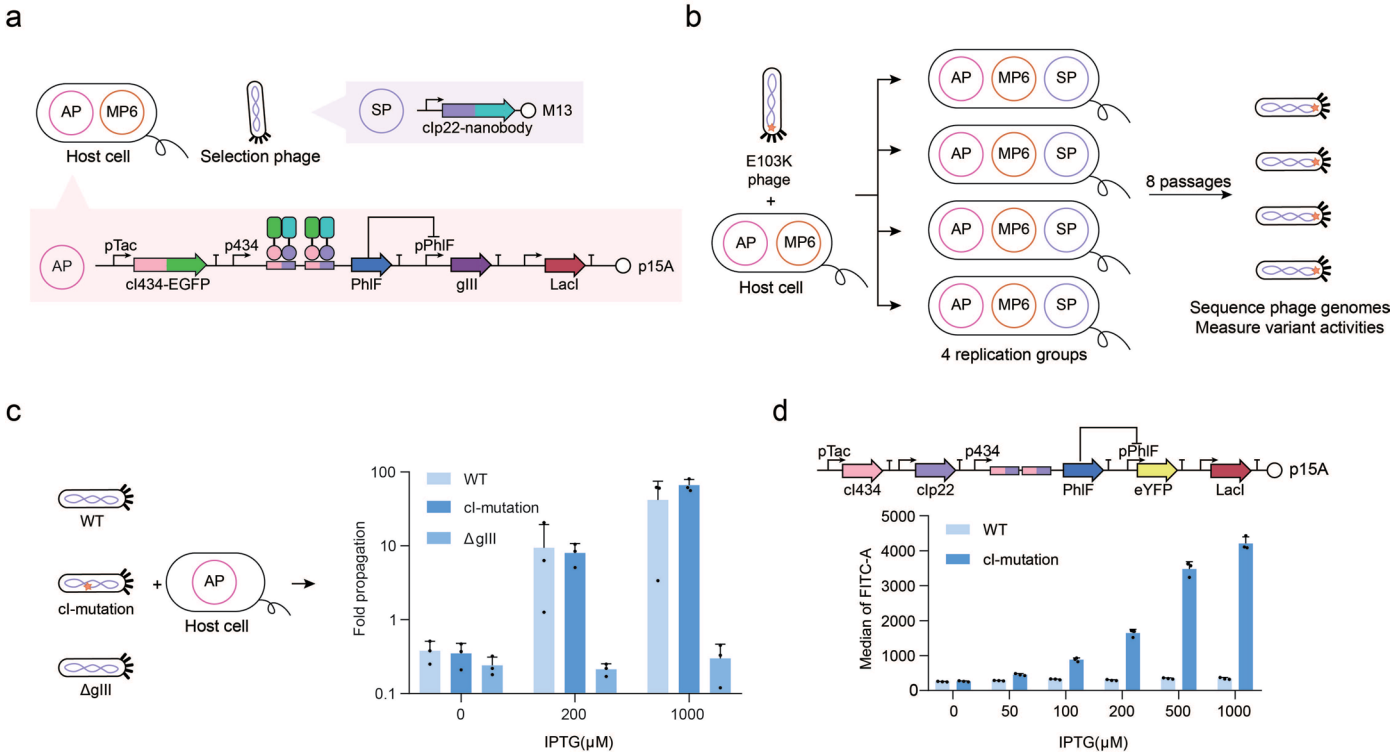
Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41592-024-02504-2.

Correspondence and requests for materials should be addressed to Shuyi Zhang.

Peer review information *Nature Methods* thanks the anonymous reviewers for their contribution to the peer review of this work. Primary Handling Editor: Arunima Singh, in collaboration with the *Nature Methods* team. Peer reviewer reports are available.

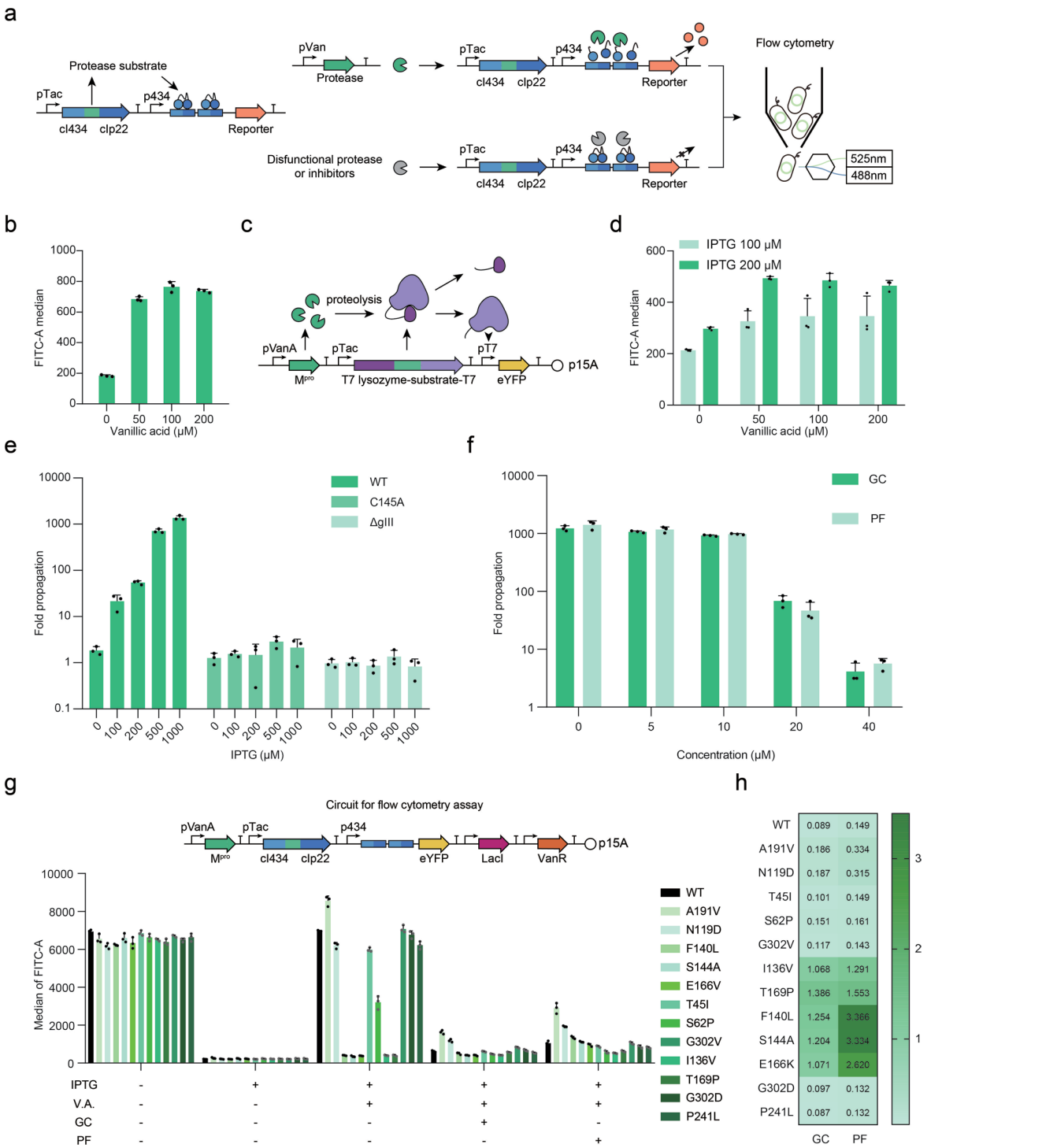
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Extended Data Fig. 1 | PANCE for EGFP-nanobody. (**a**) Genetic circuit design of PANCE for EGFP-nanobody. (**b**) Diagram of PANCE process for EGFP-nanobody. (**c**) Phage propagation assays of the cI-mutation nanobody compared to WT and ΔgIII. (**d**) Flow cytometry assays of interaction between the mutated clp22 and the WT cI434. Concentration of IPTG was 0, 50, 100, 200, 500 and 1000 μM. Data are mean ± SD of three experiments.

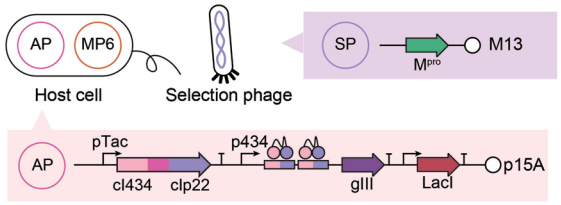
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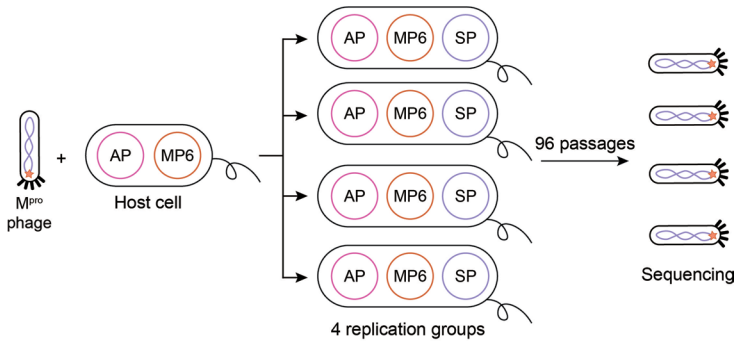
Extended Data Fig. 2 | The protease activity testing system. (**a**) Schematic diagram of reverse two-hybrid protease activity testing system. Matched substrate was used as the linker of the cI repressor pair, and the active protease cut the substrate and rescued the activity of the p434 promoter repressed by the fused cI repressor. Expression of the downstream reporters (fluorescence protein, pIII for phage propagation, etc.) was controlled by the p434 promoter. (**b**) Flow cytometry assays for HCV protease by the reverse two-hybrid protease activity testing system. Different concentrations of vanillic acid (0, 50, 100, 200 μM) and IPTG (100 μM) were used. (**c**) Schematic diagram of the T7-based protease activity testing system. T7 lysozyme and T7 polymerase were linked together with a linker carrying the protease substrate. As the protease cut the substrate, released T7 polymerase recognized T7 promoter and expressed the downstream reporter. (**d**) Flow cytometry assays for M^{pro} by T7-based protease activity system (IPTG = 100, 200 μM, Vanillic acid=0, 50, 100, 200 μM). (**e**) Phage propagation assays for M^{pro} and its variants under different concentrations of IPTG (IPTG = 0, 100, 200, 500, 1000 μM). (**f**) Phage propagation assays for inhibitors with different concentrations. Inhibitor concentration gradient was set as 0, 5, 10, 20, 40 μM. The concentration of IPTG was set as 1000 μM. (**g**) Flow cytometry assays for representative M^{pro} variants obtained by EvoScan (IPTG = 100 μM, vanillic acid=100 μM, GC376 = 50 μM, PF-07321332 = 50 μM). (**h**) Resistance Index of M^{pro} variants on both inhibitors (50 μM). Concentration of IPTG was 100 μM, and vanillic acid was 100 μM. Data are mean ± SD of three experiments.

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a



b

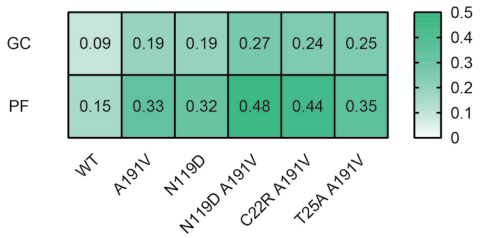


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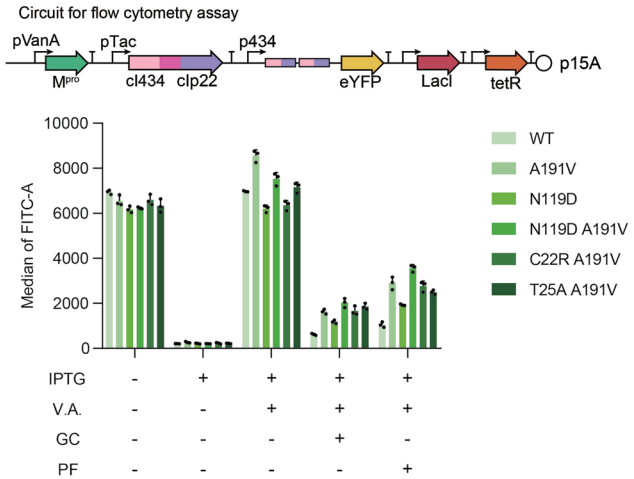
Passage	4	12	24	36	.....	96
GC376(μM)	4	4	10	10	.....	80
Group 1	NSM.	NSM.	NSM.	NSM.	.....	N119D A191V
Group 2	NSM.	NSM.	NSM.	NSM.	.....	C22R A191V
Group 3	NSM.	NSM.	NSM.	NSM.	.....	C22R A191V
Group 4	NSM.	NSM.	NSM.	NSM.	.....	T25A A191V

NSM: no siginificant mutation

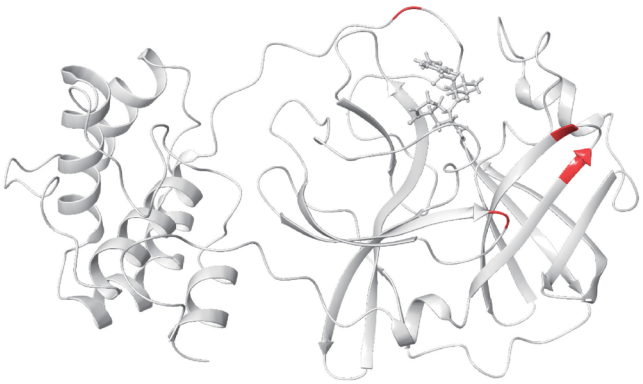
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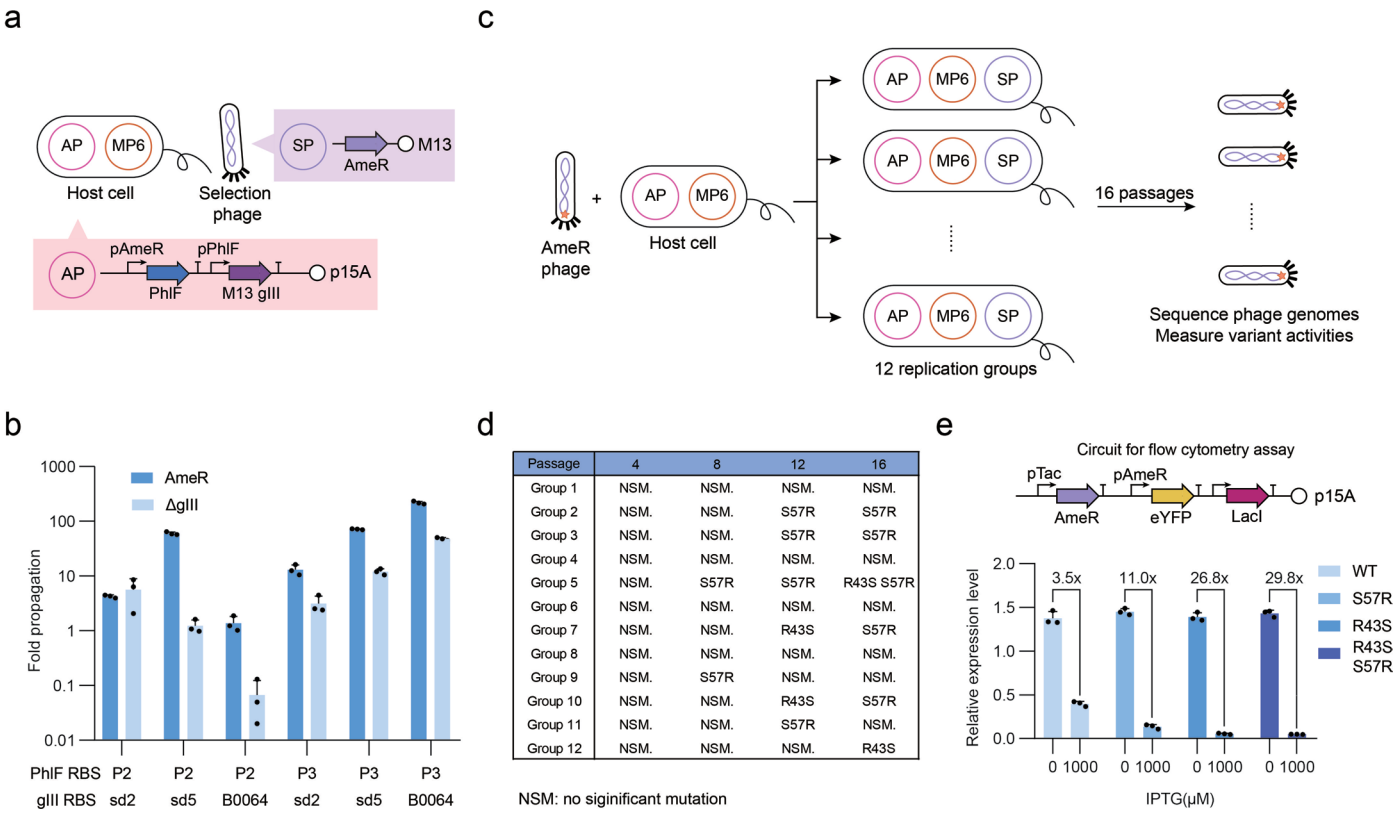


f



Extended Data Fig. 3 | PANCE to obtain M^{pro} variants that escape the inhibition of GC376. (**a**) Genetic circuit design of PANCE for M^{pro}. (**b**) Diagram of PANCE process for M^{pro}. 4 replication groups were performed. (**c**) Evolution results of PANCE after 96 passages. (**d**) Flow cytometry assays of M^{pro} variants. The A191V and N119D single mutation were also measured and served as control groups. Concentration of IPTG was 100 μM, concentration of vanillic acid was 100 μM, and concentration of GC376 was 50 μM. (**e**) Resistance Index of variants against GC376 after 96 passages. The A191V and N119D single mutation were also shown as control groups. (**f**) Mutations from PANCE mapped to the crystal structure of M^{pro} (PDB ID: 7CB7). Data are mean ± SD of three experiments.

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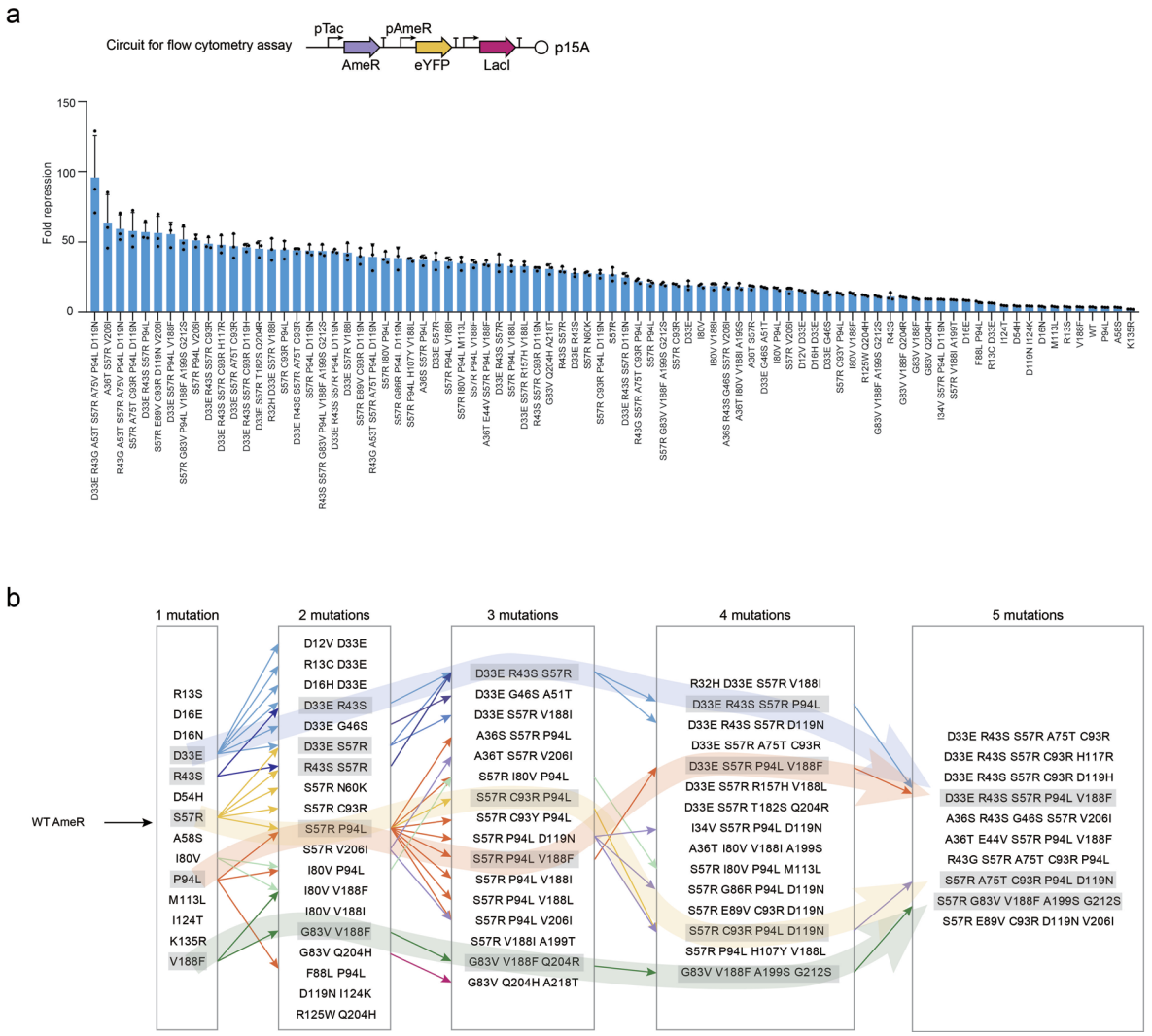


Passage	4	8	12	16
Group 1	NSM.	NSM.	NSM.	NSM.
Group 2	NSM.	NSM.	S57R	S57R
Group 3	NSM.	NSM.	S57R	S57R
Group 4	NSM.	NSM.	NSM.	NSM.
Group 5	NSM.	S57R	S57R	R43S S57R
Group 6	NSM.	NSM.	NSM.	NSM.
Group 7	NSM.	NSM.	R43S	S57R
Group 8	NSM.	NSM.	NSM.	NSM.
Group 9	NSM.	S57R	NSM.	NSM.
Group 10	NSM.	NSM.	R43S	S57R
Group 11	NSM.	NSM.	S57R	NSM.
Group 12	NSM.	NSM.	NSM.	R43S

NSM: no siginificant mutation

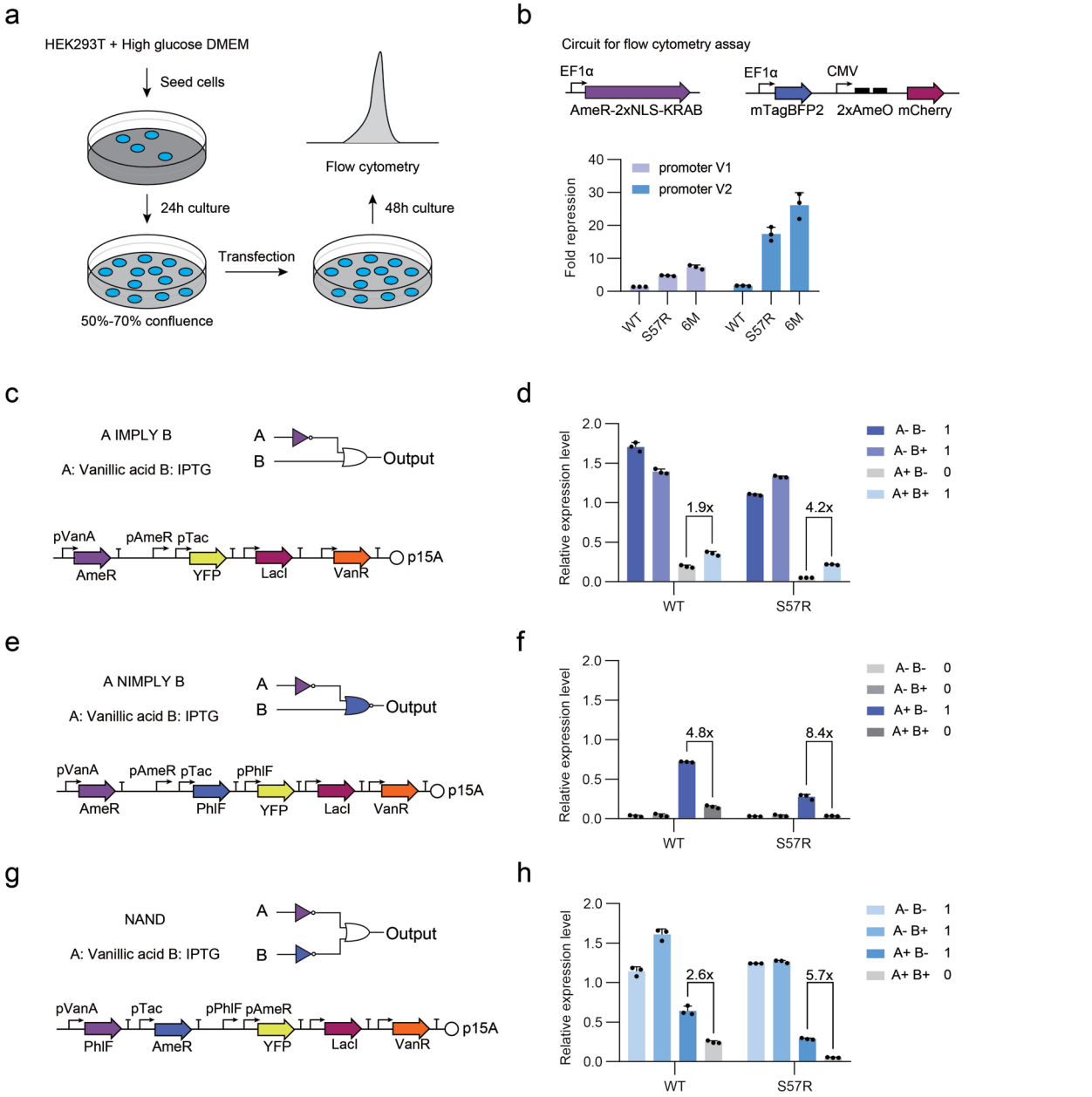
Extended Data Fig. 4 | PANCE for AmeR evolution. (**a**) Genetic circuit design of PANCE for AmeR evolution. (**b**) Propagation assays of combinatorial AP designs. Different RBS combinations for PhlF and gIII on APs were tested. (**c**) Workflow of PANCE for AmeR. 12 replication groups were performed. (**d**) Evolution results of PANCE for AmeR after 16 passages. (**e**) Flow cytometry assay of variants from PANCE for AmeR. Data are mean ± SD of three experiments.

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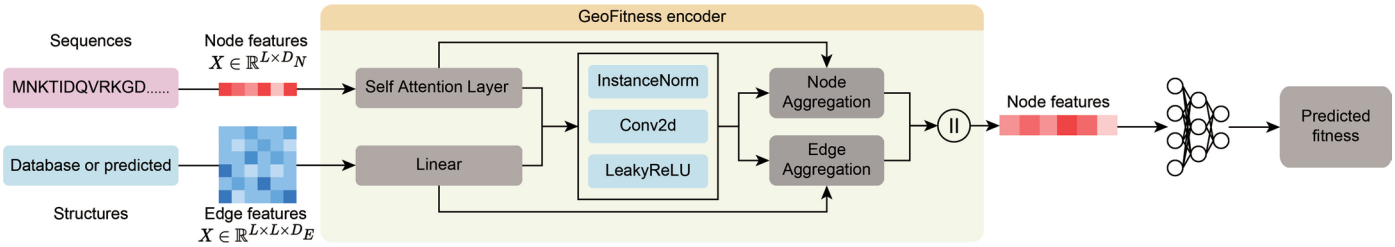
Extended Data Fig. 5 | AmeR variant activity measurement and evolution path analysis. (**a**) Fold repression of 82 AmeR variants from EvoScan. Data are mean ± SD of three experiments. The circuit for flow cytometry assay was shown. (**b**) Possible evolutionary paths that lead to the same variants with multiple mutations.

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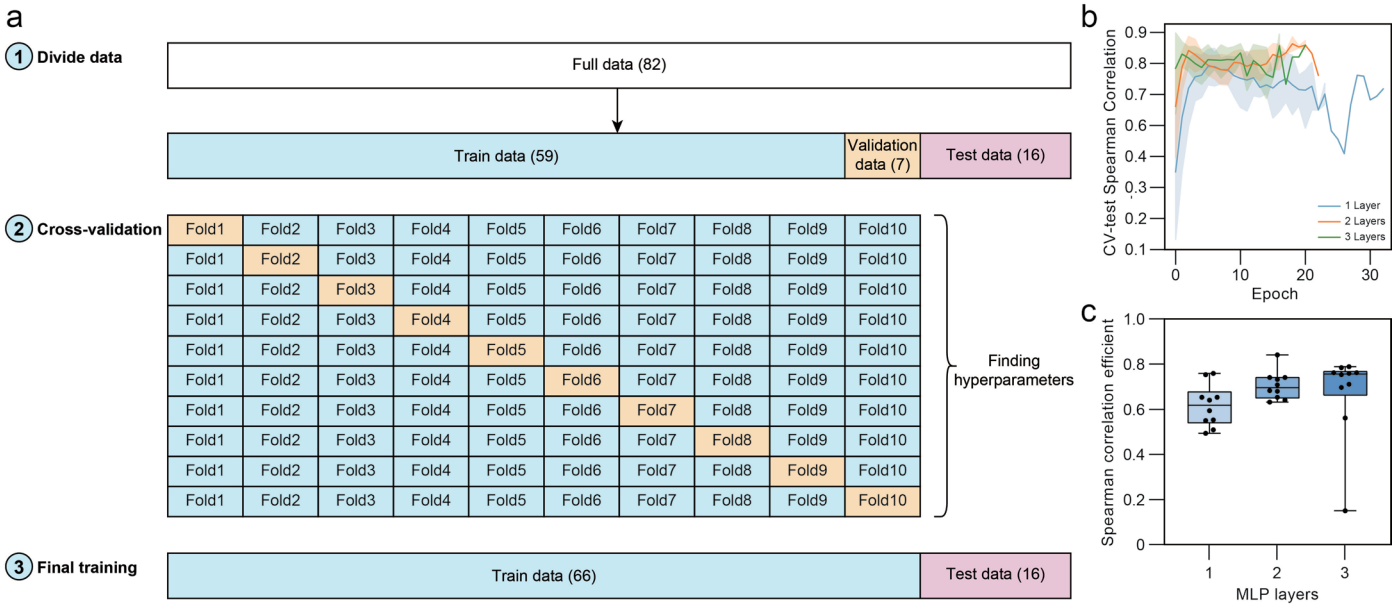
Extended Data Fig. 6 | Genetic circuit construction with WT AmeR and the S57R variant. (**a**) Schematic diagram of AmeR flow cytometry assay in mammalian cell (HEK293T). (**b**) The genetic circuit for AmeR flow cytometry assay and the results. 6 M is the AmeR variant R43G A53T S57R A75V P94L D119N. (**c**) Genetic circuit design of A IMPLY B. (**d**) Flow cytometry assay of the genetic circuit A IMPLY B. (**e**) Genetic circuit design of A NIMPLY B. (**f**) Flow cytometry assay of the genetic circuit A NIMPLY B. (**g**) Genetic circuit design of NAND. (**h**) Flow cytometry assay of the genetic circuit NAND. All circuits in this figure used 100 μM vanillic acid (input A) and 1000 μM IPTG (input B) as input signals. Data are mean ± SD of three experiments.

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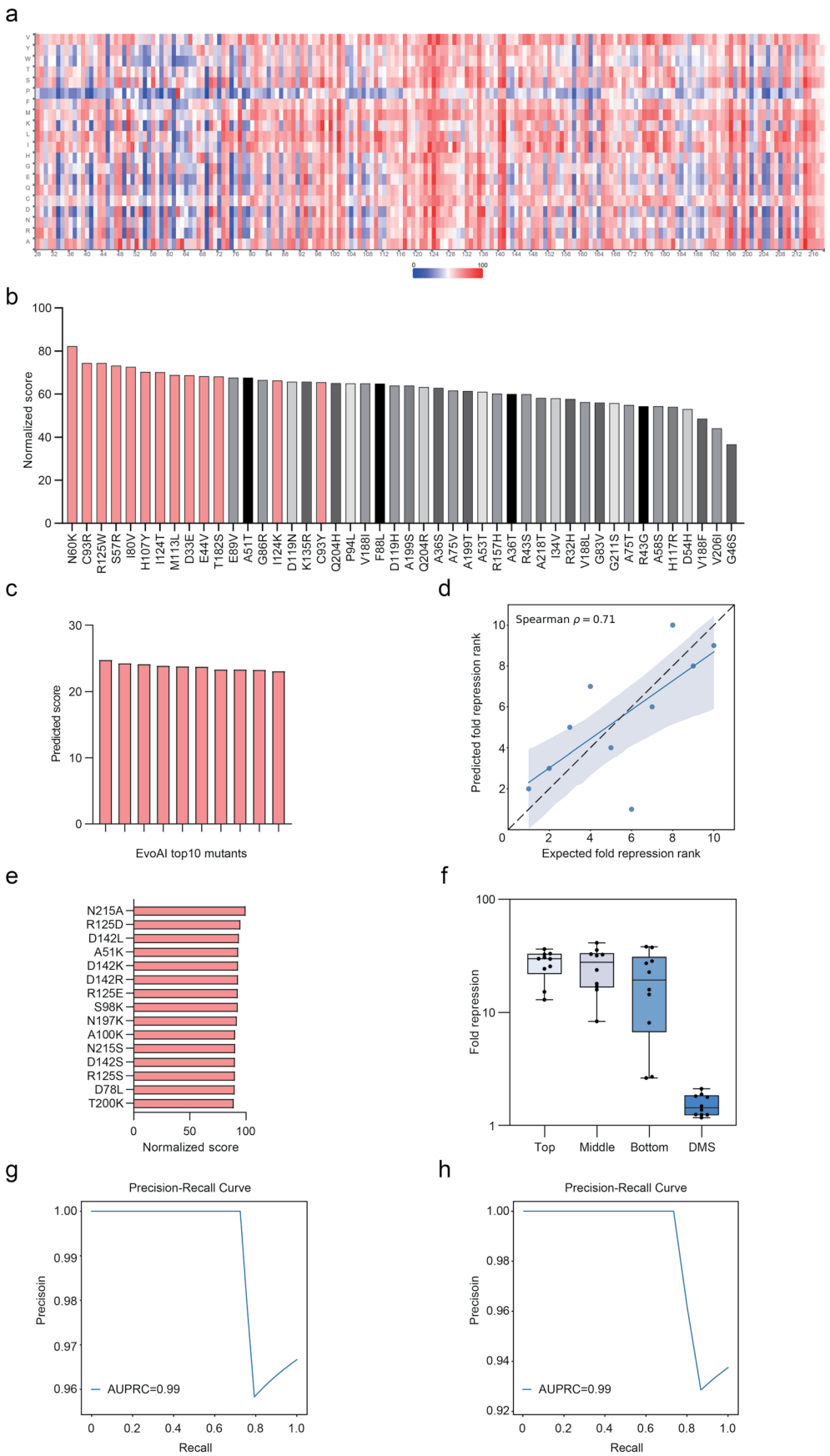
Extended Data Fig. 7 | Overall algorithm flow of the GeoFitness geometric encoder. For each protein variant, sequence information was transferred into a 1-dimensional vector, and the structure of the sample collected from PDB or predicted by Alphafold2 was transferred into a 2-dimensional vector. These two inputs were integrated through the geometric encoder block and the output was put into the Multi-Layer Perceptron (MLP) for generation of predicted fitness.

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Extended Data Fig. 8 | Cross-Validation of EvoAI model training. (**a**) Schematic diagram of the data set in 10-fold cross-validation. (**b**) CV-test spearman correlation coefficient of different layers during cross-validation. Curves are shown as the mean of 10 groups. The shadow shows the 95% confidence interval of Spearman correlation values among training process. (**c**) Influence of layer number of MLP on 10-fold cross-validation of model training. 2-layer MLP shows the best performance with higher spearman correlation coefficient compared to 1-layer MLP and smaller variance compared to 3-layer MLP. The centre line represented the median value, while the box contained a quarter to three quarters of the dataset. The minima and maxima were also shown by the whiskers.

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Extended Data Fig. 9 | See next page for caption.

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Extended Data Fig. 9 | GeoFitness values of AmeR mutations. (**a**) GeoFitness values of all single site mutations generated by the pre-trained model. The first 28 sites of N-terminal were discarded in prediction because of low confidence. A larger score indicates a higher likelihood that this mutation will improve protein function. (**b**) GeoFitness value ranking of mutations from all the anchors. The selected top 11 sites (13 mutations in total) were colored in red. (**c**) Predicted score of the top 10 variants each with a combination of 6 mutations designed by EvoAI. (**d**) Spearman correlation coefficient of the predicted fold repression rank and the experimental fold repression rank of the top 10 variants. The shaded area around the fitted line represents the 95% confidence interval. (**e**) Top 15 mutations with the highest predicted GeoFitness values from all single mutations. (**f**) Experimental fold repression of the designed variants using model trained by EvoScan anchors (Top, Middle, Bottom) or deep mutational scanning (DMS) information. Data points are mean of three biological replicates. The centre line represented the median value, while the box contained a quarter to three quarters of the dataset. The minima and maxima were also shown by the whiskers. (**g**) AUPRC plot for EvoAI-generated variants. (**h**) AUPRC plot for the test set during EvoAI training.

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☒	☐	For null hypothesis testing, the test statistic (e.g. *F*, *t*, *r*) with confidence intervals, effect sizes, degrees of freedom and *P* value noted *Give P values as exact values whenever suitable.*
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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	Flow cytometry data were collected by CytExpert 2.4 through flow cytometer (Beckman Coulter Cytoflex S).
Data analysis	Flow cytometry data were analyzed by FlowJo v10; Protein sequences of the 82 variants and the WT were collected as a fasta file and the file was input into MEGA11 for multiple sequence alignment (MSA). After MSA and phylogenetic analysis, neighbor-joining tree was selected as the method of tree construction. The output tree was decorated by iTOL(v6.9.1), and all the parameters were set using default values. Results were presented by Graphpad Prism 8. For EvoAI feature generation, we employ AlphaFold2 to construct the structural representation of the wild-type protein. For this endeavor, we deploy the GeoFitness-Seq variant of the pre-training model. In the case of mutated proteins, structural configurations are generated using FoldX 5. The sequence features are extracted from the large-scale protein language model ESM-2, for the purpose of capturing global context information. Original codes for EvoAI could be freely accessed at https://github.com/Gonglab-THU/EvoAI.

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All manuscripts must include a data availability statement. This statement should provide the following information, where applicable:

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Data for phage propagation assay, flow ctyometry assay and EvoAI training and test are accessible in the source data. Other data and materials used in this study are available from the corresponding authors by reasonable request. PDB files used in this study include PDB ID: 7CB7 and PDB ID: 7VLO. All the prediction features and results of mutants are available at https://zenodo.org/doi/10.5281/zenodo.10686156.

Human research participants

Policy information about studies involving human research participants and Sex and Gender in Research.

Reporting on sex and gender	N/A
Population characteristics	N/A
Recruitment	N/A
Ethics oversight	N/A

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size	For variant study, sample size was determined based on the directed evolution data. We calculated all the mutants we captured and set the total number of the variants as our sample size.
Data exclusions	No data were excluded.
Replication	Excepting the evolution passages (replications are not necessarily needed) , all experiments had at least three replications.
Randomization	No randomization was used. For evolution process we didn't randomize or simulate pseudo data for further deep learning. Therefore we didn't use randomization process in this study.
Blinding	No blinding was performed. There's no need to setup blinding control group in the directed evolution process of this study. All cells and bacteria were prepared under standard protocol.

Reporting for specific materials, systems and methods

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Materials & experimental systems

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☒	☐ Antibodies
☐	☒ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☐	☒ Flow cytometry
☒	☐ MRI-based neuroimaging

Eukaryotic cell lines

Policy information about cell lines and Sex and Gender in Research

Cell line source(s)	Cell line HEK293T was retrieved from ATCC.
Authentication	Cell line HEK293T was not authenticated by us. We purchased it commercially.
Mycoplasma contamination	Cell line was regularly tested negative for mycoplasma.
Commonly misidentified lines (See ICLAC register)	No commonly misidentified lines were used in this study.

Flow Cytometry

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Methodology

Sample preparation	For E.coli, 10 µl of the bacteria was added into 190 µl PBS with 2 g/L kanamycin in the 96-well U-bottom plate to stop the cell growth. For mammalian cells HEK293T, cells were trypsinized 48 h after transfection and were then centrifuged at 250 × g for 10 min at room temperature. The supernatant was removed, and the cells were resuspended in 1 × PBS.
Instrument	The flow cytometer (Beckman Coulter Cytoflex S) was used to quantify the expression levels of fluorescent protein.
Software	The flow cytometry data was collected using CytExpert 2.4 and analyzed with FlowJo v10.
Cell population abundance	For E.coli, at least 10000 events were collected within 60 seconds and 60 µL for each sample; For mammalian cells, the median of fluorescence was calculated for >20,000 transfected cells for each sample.
Gating strategy	For E.coli, live cells were gated using FSC-A/SSC-A. For Mammalian cells, Data were gated by the area of the forward scatter and the side scatter (FSC-A/SSC-A) and then cell populations were selected by gating out the background BFP signal of untransfected cells to obtain the median of fluorescence.

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